MATH 521

Real Analysis

Final Practice Book

Write-On Workbook

University of Wisconsin–Madison

Spring 2026

Current build: Homework 5 through Homework 12

Name: ____________________

Date: ____________________

Contents

How to Use This Workbook

This workbook is for active proof practice. For each problem, work through the diagnosis, recipe card, self-explanation, skeleton, and full proof attempt before reading the reference solution.

Pass 1 Diagnose the topic, task, relevant definitions, and likely proof method.

Pass 2 Convert the problem into hypotheses, conclusion, tools, and a proof plan.

Pass 3 Answer targeted questions that expose the proof's logic.

Pass 4 Reconstruct the proof from a skeleton without reading the full solution.

Pass 5 Write a complete proof from scratch in the blank writing space.

Only after Pass 5 should you compare your work to the official or reference solution.

Homework 5

Problem 1

Problem Statement **Homework 5, Problem 1**

Let $\{a_n\}_{n\geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n\to\infty} a_n = +\infty$ if and only if $\lim_{n\to\infty} \frac{1}{a_n} = 0$.

Pass 1 – First Diagnosis **Homework 5, Problem 1**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 5, Problem 1

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 5, Problem 1**

1. What is the definition of $a_n \to +\infty$?

2. Why is positivity of a_n essential before taking reciprocals?

3. In the forward direction, what should M be in terms of ε?

4. In the reverse direction, what should ε be in terms of M?

5. Where does the inequality reversal happen?

Pass 4 – Skeleton Proof **Homework 5, Problem 1**

Forward direction.
Fix $\varepsilon > 0$. Use $a_n \to +\infty$ with $M =$

For all $n \geq N$, take reciprocals to obtain:

Conclude $\left|\frac{1}{a_n} - 0\right| < \varepsilon$:

Reverse direction.
Fix $M > 0$. Use $\frac{1}{a_n} \to 0$ with $\varepsilon =$

Use positivity to convert $\frac{1}{a_n} < \frac{1}{M}$ into:

State the definition this verifies:

Pass 5 – Full Proof Attempt Homework 5, Problem 1

Use this page for both directions.

Official Solution (Professor)

$\Rightarrow$ Fix $\varepsilon > 0$. As $\lim a_n = +\infty$, then there exists N so that

$$a_n > \frac{1}{\varepsilon} \qquad \text{for all } n \geq N.$$

So, for $n \geq N$ we have

$$-\varepsilon < 0 < \frac{1}{a_n} < \varepsilon \qquad \Longrightarrow \qquad \left|\frac{1}{a_n} - 0\right| < \varepsilon.$$

Therefore $\lim \frac{1}{a_n} = 0$.
$\Leftarrow$ Fix $M > 0$. As $\lim \frac{1}{a_n} = 0$, then there exists N so that

$$\left|\frac{1}{a_n} - 0\right| < \frac{1}{M} \qquad \text{for all } n \geq N.$$

Note that $\left|\frac{1}{a_n} - 0\right| = \frac{1}{a_n} > 0$. So

$$a_n > M \qquad \text{for all } n \geq N.$$

Therefore $\lim a_n = +\infty$.

Reflection & Variation **Homework 5, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 5, Problem 2**

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that if $\{a_{k_n}\}_{n\geq 1}$ is a subsequence of $\{a_n\}_{n\geq 1}$ that converges to $L \in \mathbb{R}$, then

$$\liminf_{n\to\infty} a_n \leq L \leq \limsup_{n\to\infty} a_n.$$

(Hint: As $\{a_n\}_{n\geq 1}$ is not necessarily bounded, we have to consider the cases $\liminf_{n\to\infty} a_n = \pm\infty$ and $\limsup_{n\to\infty} a_n = \pm\infty$. However, the proof is much easier in these cases.)

Pass 1 – First Diagnosis **Homework 5, Problem 2**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 5, Problem 2**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 5, Problem 2**

1. What do $\limsup a_n$ and $\liminf a_n$ measure using tail suprema/infima?

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2. Why does $k_n \geq n$ imply that a fixed tail eventually contains the subsequence terms?

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3. Why is the case $\limsup a_n = +\infty$ immediate?

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4. Why is the case $\limsup a_n = -\infty$ impossible when $L \in \mathbb{R}$?

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5. How do you get $a_{k_n} \leq \sup\{a_m : m \geq N\}$ for all large n?

Pass 4 – Skeleton Proof **Homework 5, Problem 2**

Prove $L \le \limsup a_n$.
Separate the easy cases $\limsup a_n = +\infty$ and $\limsup a_n = -\infty$:

Assume $\limsup a_n \in \mathbb{R}$. For fixed N, compare a_{k_n} to the tail supremum for all large n:

Pass $n \to \infty$ to get $L \le \sup\{a_m : m \ge N\}$:

Pass $N \to \infty$ to obtain $L \le \limsup a_n$:

Prove $\liminf a_n \le L$.
Repeat the same argument using tail infima and reverse inequalities:

Pass 5 – Full Proof Attempt | Homework 5, Problem 2

Use this page for the $\limsup$ **inequality.**

Pass 5 – Full Proof Attempt **Homework 5, Problem 2**

Use this page for the $\liminf$ **inequality and endpoint cases.**

Official Solution (Professor)

Let's start with the second inequality:

$$L \leq \limsup_{n\to\infty} a_n.$$

Case: $\limsup a_n = +\infty$. Then the statement is automatically true, since every real number L satisfies $L < +\infty$.
Case: $\limsup a_n = -\infty$. As $\liminf a_n \leq \limsup a_n$, then we must have $\liminf a_n = -\infty = \limsup a_n$. This means that $\lim a_n = -\infty$ by Lecture 14.
Fix $m < 0$. As $\lim a_n = -\infty$, there exists N so that

$$a_n < m \qquad \text{for all } n \geq N.$$

As $k_n \geq n$ for all n, this implies

$$a_{k_n} < m \qquad \text{for all } n \geq N.$$

Therefore $L = \lim a_{k_n} = -\infty$. This contradicts that $L \in \mathbb{R}$, and so this case is impossible.
Case: $\limsup a_n \in \mathbb{R}$. For $N \geq 1$, notice that

$$a_{k_n} \leq \sup\{a_n : n \geq N\} \qquad \text{for all } n \geq N.$$

We know that the sequence $\{a_{k_n}\}$ converges to L. So, by HW4#3, we have

$$L = \lim a_{k_n} \leq \sup\{a_n : n \geq N\}.$$

On the other hand, $v_N = \sup\{a_n : n \geq N\}$ is a sequence that converges to $\limsup a_n$. So, using HW4#3 again, we conclude that

$$L \leq \lim_{N\to\infty} \sup\{a_n : n \geq N\} = \limsup a_n.$$

Exercise: Using similar steps, prove that

$$\liminf_{n\to\infty} a_n \leq L.$$

Reflection & Variation **Homework 5, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 5, Problem 3**

Let $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ be two bounded sequences. Prove that

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Pass 1 – First Diagnosis **Homework 5, Problem 3**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 5, Problem 3

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 5, Problem 3**

1. Why does boundedness guarantee finite tail suprema for a_n, b_n, and $a_n + b_n$?

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2. For a fixed N, what upper bounds do the tail suprema give for a_n and b_n?

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3. Why does adding two pointwise upper bounds produce an upper bound for $a_n + b_n$?

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4. How does an upper bound for every tail term become an inequality of suprema?

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5. Which limit law finishes the proof after passing $N \to \infty$?

Pass 4 – Skeleton Proof **Homework 5, Problem 3**

Fix $N \geq 1$. Write the two tail upper bounds for all $n \geq N$:

Add them to get an upper bound for $a_n + b_n$:

Take the supremum over $n \geq N$ to obtain:

Let $N \to \infty$ and identify each term with a limsup:

State the final inequality:

Use this page for the complete proof.

Official Solution (Professor)

Fix $N \geq 1$. For $n \geq N$ we have

$$a_n \leq \sup\{a_n : n \geq N\}, \qquad b_n \leq \sup\{b_n : n \geq N\},$$

and so

$$a_n + b_n \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

In other words, $\sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}$ is an upper bound for the set $\{a_n + b_n : n \geq N\}$. Therefore

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

Altogether, we have shown that for any $N \geq 1$ we have

$$\sup\{a_n + b_n : n \geq N\} \leq \sup\{a_n : n \geq N\} + \sup\{b_n : n \geq N\}.$$

As $\{a_n\}$ is a bounded sequence, we know from lecture that the sequence $v_N = \sup\{a_n : n \geq N\}$ converges to $v = \limsup a_n \in \mathbb{R}$. Similarly, as $\{b_n\}$ is bounded, the sequence $\sup\{b_n : n \geq N\}$ converges to $\limsup b_n$ as $N \to \infty$. By the inequality above, we see that the sequence $\sup\{a_n + b_n : n \geq N\}$ is also bounded above. Therefore $\limsup(a_n + b_n)$ is a finite number, and $\sup\{a_n + b_n : n \geq N\}$ converges to $\limsup(a_n + b_n)$ as $N \to \infty$.

By the limit laws, the sequence

$$\sup\{a_n + b_n : n \geq N\} - \sup\{a_n : n \geq N\} - \sup\{b_n : n \geq N\}$$

converges as $N \to \infty$. We also know that it is ≤ 0 for all $N \geq 1$. Therefore, by HW4#3, the limit is also ≤ 0:

$$\limsup(a_n + b_n) - \limsup a_n - \limsup b_n \leq 0$$

and hence

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Reflection & Variation **Homework 5, Problem 3**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 5, Problem 4**

Let $\{a_n\}_{n\geq 1}$ be a sequence. Prove that $\limsup_{n\to\infty} |a_n| = 0$ if and only if $\lim_{n\to\infty} a_n = 0$.

Pass 1 – First Diagnosis **Homework 5, Problem 4**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 5, Problem 4**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 5, Problem 4**

1. Why is $\liminf |a_n| \geq 0$ **automatic?**

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2. How does $\liminf |a_n| = \limsup |a_n| = 0$ **imply** $|a_n| \to 0$**?**

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3. Why is $|a_n| \to 0$ **exactly the same as** $a_n \to 0$**?**

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4. In the reverse direction, which earlier homework result gives $|a_n| \to 0$**?**

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5. How does Lecture 14 connect ordinary limits with liminf and limsup?

Pass 4 – Skeleton Proof **Homework 5, Problem 4**

Forward direction.
Assume $\limsup |a_n| = 0$**. Prove** $\liminf |a_n| \geq 0$**:**

Use $\liminf |a_n| \leq \limsup |a_n|$ **to conclude:**

Invoke the liminf/limsup convergence criterion to get $|a_n| \to 0$**:**

Translate $|a_n| \to 0$ **into** $a_n \to 0$**:**

Reverse direction.
Assume $a_n \to 0$**. Use HW4#1 to get:**

Use Lecture 14 to identify $\limsup |a_n|$:

Use this page for both directions.

Official Solution (Professor)

$\Rightarrow$ Suppose $\limsup |a_n| = 0$.
Claim: $\liminf |a_n| \geq 0$. As $|a_n| \geq 0$ for all n, we have

$$\inf_{n \geq N} |a_n| \geq 0 \qquad \text{for all } N \geq 1.$$

Therefore

$$\lim_{N\to\infty} \inf_{n\geq N} |a_n| \geq 0$$

by HW4#3.
Altogether, we have

$$0 \leq \liminf |a_n| \leq \limsup |a_n| = 0.$$

Therefore

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

By Lecture 10, this means that

$$\lim |a_n| = 0.$$

In other words: for all $\varepsilon > 0$ there exists N so that

$$||a_n| - 0| < \varepsilon \qquad \text{for all } n \geq N.$$

This is the same as: for all $\varepsilon > 0$ there exists N so that

$$|a_n - 0| < \varepsilon \qquad \text{for all } n \geq N.$$

So $\lim a_n = 0$.
$\Leftarrow$ Suppose $\lim a_n = 0$. By HW4#1, this implies

$$\lim |a_n| = 0.$$

By Lecture 14, we have

$$\liminf |a_n| = 0 = \limsup |a_n|.$$

So $\limsup |a_n| = 0$, as desired.

Reflection & Variation **Homework 5, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 6

Problem 1

Problem Statement **Homework 6, Problem 1**

Prove that the series
$$\sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$
converges, and find its value. (Hint: You computed the partial sums on Homework 1.)

Pass 1 – First Diagnosis **Homework 6, Problem 1**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 6, Problem 1**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 6, Problem 1**

1. What does it mean for a series to converge in terms of partial sums?

2. What partial fraction decomposition should be used?

3. Which terms cancel in the Nth partial sum?

4. What explicit formula for S_N should remain after cancellation?

5. What limit gives the value of the series?

Pass 4 – Skeleton Proof **Homework 6, Problem 1**

Define the Nth partial sum $S_N =$

Rewrite the summand using partial fractions:

Write the telescoping sum and identify the surviving first and last terms:

Compute $\lim_{N\to\infty} S_N$:

State convergence and the value of the series:

Use this page for the complete proof.

Student-Derived Solution (No official solution available)

Let

$$a_n = \frac{1}{(3n-2)(3n+1)}$$

and let $S_N = \sum_{n=1}^{N} a_n$. Using partial fractions,

$$\frac{1}{(3n-2)(3n+1)} = \frac{1}{3}\left(\frac{1}{3n-2} - \frac{1}{3n+1}\right).$$

Hence

$$S_N = \frac{1}{3}\sum_{n=1}^{N}\left(\frac{1}{3n-2} - \frac{1}{3n+1}\right).$$

Writing out the terms gives

$$S_N = \frac{1}{3}\left[\left(1-\frac{1}{4}\right) + \left(\frac{1}{4}-\frac{1}{7}\right) + \cdots + \left(\frac{1}{3N-2} - \frac{1}{3N+1}\right)\right].$$

All intermediate terms cancel, so

$$S_N = \frac{1}{3}\left(1 - \frac{1}{3N+1}\right) = \frac{N}{3N+1}.$$

Therefore

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty}\frac{N}{3N+1} = \frac{1}{3}.$$

Since the sequence of partial sums converges, the series converges, and its value is

$$\sum_{n=1}^{\infty}\frac{1}{(3n-2)(3n+1)} = \frac{1}{3}.$$

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 6, Problem 2**

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be convergent series.

(a) Prove that for any $k \in \mathbb{R}$, the series $\sum_{n=1}^{\infty} ka_n$ converges and

$$\sum_{n=1}^{\infty} ka_n = k \sum_{n=1}^{\infty} a_n.$$

(b) Prove that the series $\sum_{n=1}^{\infty}(a_n + b_n)$ converges and

$$\sum_{n=1}^{\infty}(a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Pass 1 – First Diagnosis **Homework 6, Problem 2**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 6, Problem 2

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 6, Problem 2**

1. How is the sum of a convergent series defined using partial sums?

---

2. In part (a), what are the partial sums of $\sum ka_n$ in terms of the partial sums of $\sum a_n$?

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3. Which sequence limit law proves convergence in part (a)?

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4. In part (b), how do the partial sums of $\sum(a_n + b_n)$ split?

---

5. Which limit law gives the final identity for the sum?

Pass 4 – Skeleton Proof **Homework 6, Problem 2**

Part (a).
Let $s_n = \sum_{j=1}^{n} a_j$. The partial sums of $\sum k a_n$ are:

Use $s_n \to \sum a_n$ and the constant multiple limit law:

Translate the limit of partial sums into the series identity:

Part (b).
Let $s_n = \sum_{j=1}^{n} a_j$ and $t_n = \sum_{j=1}^{n} b_j$. The partial sums are:

Use the addition limit law and state the resulting sum:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Let $s_n = \sum_{j=1}^{n} a_j$ denote the nth partial sum. As $\sum a_n$ converges, we know that the sequence $\{s_n\}$ converges.

Fix $k \in \mathbb{R}$. Then the partial sums $\sum_{j=1}^{n} ka_j$ are equal to ks_n. By the limit law for multiplication by a constant, we conclude that the sequence $\{ks_n\}_{n\geq 1}$ converges and $\lim(ks_n) = k \lim s_n$. By definition, this is exactly what it means for the series $\sum ka_n$ to converge to the number $k \sum a_n$.

(b) Let $s_n = \sum_{j=1}^{n} a_j$ and $t_n = \sum_{j=1}^{n} b_j$ denote the nth partial sums. As $\sum a_n$ and $\sum b_n$ converge, we know that the sequences $\{s_n\}$ and $\{t_n\}$ converge.

Note that the partial sums $\sum_{j=1}^{n}(a_j + b_j)$ are equal to $s_n + t_n$. By the limit law for addition, we conclude that the sequence $\{s_n + t_n\}_{n\geq 1}$ converges and $\lim(s_n + t_n) = \lim s_n + \lim t_n$. By definition, this is exactly what it means for the series $\sum(a_n + b_n)$ to converge to the number $\sum a_n + \sum b_n$.

Reflection & Variation | **Homework 6, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 6, Problem 3**

(a) Prove that
$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$
(Hint: $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$.)

(b) Use part (a) to compute
$$\sum_{n=1}^{\infty} \frac{n}{2^n}.$$

Pass 1 – First Diagnosis Homework 6, Problem 3

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 6, Problem 3

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 6, Problem 3**

1. What identity turns the summand in part (a) into a telescoping difference?

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2. What formula should the nth partial sum satisfy?

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3. Why does $\frac{n}{2^n} \to 0$?

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4. In part (b), how do you rewrite $\frac{n}{2^n}$ using $\frac{n-1}{2^{n+1}}$ and $\frac{1}{2^n}$?

---

5. Which geometric series value is needed?

Part (a).
Define $s_n = \sum_{k=1}^{n} \frac{k-1}{2^{k+1}}$**. Prove by induction that** $s_n =$

Base case $n = 1$**:**

Inductive step using $\frac{k-1}{2^{k+1}} = \frac{k}{2^k} - \frac{k+1}{2^{k+1}}$**:**

Pass to the limit using $\frac{n}{2^n} \to 0$**:**

Part (b).
Rewrite $\sum_{k=1}^{n} \frac{k}{2^k}$ **as** $2\sum \frac{k-1}{2^{k+1}} + \sum \frac{1}{2^k}$**:**

Use part (a) and the geometric series to compute the value:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Let $s_n = \sum_{k=1}^{n} \frac{k-1}{2^{k+1}}$.
We will prove that

$$s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}} \qquad \text{for } n \geq 1$$

by induction. For the base case, we note that

$$s_1 = \frac{0}{4} = \frac{1}{2} - \frac{2}{4}.$$

For the inductive step, assume that $s_n = \frac{1}{2} - \frac{n+1}{2^{n+1}}$ for some $n \geq 1$. Then

$$s_{n+1} = s_n + \frac{n}{2^{n+2}} = \frac{1}{2} - \frac{n+1}{2^{n+1}} + \frac{n+1}{2^{n+1}} - \frac{n+2}{2^{n+2}} = \frac{1}{2} - \frac{n+2}{2^{n+2}}.$$

Together, we conclude that our formula for s_n holds for all $n \geq 1$ by induction.
Next, we claim that $\lim \frac{n}{2^n} = 0$. Notice that $n \leq (3/2)^n$ for all $n \geq 1$, and so

$$0 \leq \frac{n}{2^n} \leq \frac{(3/2)^n}{2^n} = \left(\frac{3}{4}\right)^n.$$

The right-hand side converges to zero as $n \to \infty$. Therefore, $\frac{n}{2^n}$ also converges to zero by the squeeze theorem.
Altogether, we conclude

$$\lim s_n = \lim \left(\frac{1}{2} - \frac{n+1}{2^{n+1}}\right) = \frac{1}{2}.$$

(In the last equality, we noted that $\frac{n+1}{2^{n+1}}$ is a subsequence of $\frac{n}{2^n}$, and therefore also converges to zero.) By definition of a convergent series, we conclude that

$$\sum_{n=1}^{\infty} \frac{n-1}{2^{n+1}} = \frac{1}{2}.$$

(b) Note that

$$\sum_{k=1}^{n} \frac{k}{2^k} = 2\sum_{k=1}^{n} \frac{k-1}{2^{k+1}} + \sum_{k=1}^{n} \frac{1}{2^k}.$$

By part (a), $\sum \frac{k-1}{2^{k+1}}$ converges to $\frac{1}{2}$, and thus $2\sum \frac{k-1}{2^{k+1}}$ converges to 1 by problem 2. From lecture, we know that the geometric series $\sum \frac{1}{2^k}$ converges to $\frac{1}{1-(1/2)} - 1 = 1$. (Notice that we are starting at $k = 1$, not $k = 0$.) Together, we conclude $\sum_{k=1}^{\infty} \frac{k}{2^k}$ converges to $1 + 1 = 2$.

Reflection & Variation **Homework 6, Problem 3**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 6, Problem 4**

Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely and $\{b_n\}_{n\geq 1}$ is a bounded sequence, then the series $\sum_{n=1}^{\infty} a_n b_n$ also converges. (Hint: Use the Cauchy criterion.)

Pass 1 – First Diagnosis **Homework 6, Problem 4**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 6, Problem 4**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 6, Problem 4**

1. What does absolute convergence of $\sum a_n$ say about $\sum |a_n|$?

2. What bound does boundedness of $\{b_n\}$ provide?

3. What is the Cauchy criterion for convergence of a series?

4. How can $|\sum_{k=m}^{n} a_k b_k|$ be controlled by $\sum_{k=m}^{n} |a_k|$?

5. Why is $M + 1$ used in the denominator of the tail estimate?

Pass 4 – Skeleton Proof **Homework 6, Problem 4**

Choose $M \geq 0$ such that $|b_n| \leq M$ for all n:

Use the Cauchy criterion for $\sum |a_n|$ with tolerance $\varepsilon/(M+1)$:

For $n \geq m \geq N$, estimate $|\sum_{k=m}^{n} a_k b_k|$:

Conclude the Cauchy criterion for $\sum a_n b_n$:

Pass 5 – Full Proof Attempt Homework 6, Problem 4

Use this page for the complete proof.

Student-Derived Solution (No official solution available)

Since $\sum_{n=1}^{\infty} a_n$ converges absolutely, the series

$$\sum_{n=1}^{\infty} |a_n|$$

converges. By the Cauchy criterion for series, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\sum_{k=m}^{n} |a_k| < \frac{\varepsilon}{M+1} \qquad \text{for all } n \geq m \geq N,$$

where $M \geq 0$ is chosen so that

$$|b_n| \leq M \qquad \text{for all } n \in \mathbb{N}.$$

Such an M exists because $\{b_n\}$ is bounded.
Now let $n \geq m \geq N$. Then

$$\left| \sum_{k=m}^{n} a_k b_k \right| \leq \sum_{k=m}^{n} |a_k b_k| = \sum_{k=m}^{n} |a_k||b_k| \leq M \sum_{k=m}^{n} |a_k|.$$

Therefore

$$\left| \sum_{k=m}^{n} a_k b_k \right| < M \cdot \frac{\varepsilon}{M+1} < \varepsilon.$$

Thus, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left| \sum_{k=m}^{n} a_k b_k \right| < \varepsilon \qquad \text{for all } n \geq m \geq N.$$

By the Cauchy criterion for series, $\sum_{n=1}^{\infty} a_n b_n$ converges.

Reflection & Variation	Homework 6, Problem 4

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement **Homework 6, Problem 5**

(a) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then the series $\sum_{n=1}^{\infty} a_n^2$ also converges.

(b) Provide an example where $\sum_{n=1}^{\infty} a_n$ diverges and $\sum_{n=1}^{\infty} a_n^2$ converges.

Pass 1 – First Diagnosis **Homework 6, Problem 5**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 6, Problem 5**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 6, Problem 5**

1. Why does absolute convergence imply $a_n \to 0$?

---

2. Why does $a_n \to 0$ imply $|a_n| < 1$ eventually?

---

3. For $|x| < 1$, why is $x^2 \leq |x|$?

---

4. Which comparison proves the tail of $\sum a_n^2$ converges?

---

5. What standard example makes $\sum a_n$ diverge but $\sum a_n^2$ converge?

Pass 4 – Skeleton Proof **Homework 6, Problem 5**

Part (a).
From absolute convergence, write the convergent positive series:

Use convergence of $\sum a_n$ to get $a_n \to 0$, then choose N with $|a_n| < 1$:

For $n \geq N$, compare a_n^2 with $|a_n|$:

Use the comparison test and finite initial terms:

Part (b).
Choose $a_n =$ and cite the harmonic and p-series facts:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Note that $x^2 \leq |x|$ for any $x \in [-1, 1]$. (I recommend that you draw the graphs of x^2 and $|x|$ to check that this is true.)
As $\sum a_n$ converges, then we have $\lim a_n = 0$, and so there exists an N so that $|a_n| < 1$ for all $n \geq N$. By the previous paragraph, this means that $a_n^2 \leq |a_n|$ for all $n \geq N$. As $\sum |a_n|$ converges, then $\sum_{n=N}^{\infty} a_n^2$ also converges by the comparison test. Adding the first $N - 1$ terms back in, we conclude $\sum_{n=1}^{\infty} a_n^2$ converges as well.
(b) Consider $a_n = \frac{1}{n}$. In lecture, we proved that $\sum \frac{1}{n}$ diverges and $\sum \frac{1}{n^2}$ converges.

Reflection & Variation **Homework 6, Problem 5**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement **Homework 6, Problem 6**

For each of the following series, determine if it converges and prove your answer.

(a) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{[n+(-1)^n]^2}$ (b) $\displaystyle\sum_{n=1}^{\infty} \left(\sqrt{n+1}-\sqrt{n}\right)$ (c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n$

Pass 1 – First Diagnosis **Homework 6, Problem 6**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 6, Problem 6**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 6, Problem 6**

1. In part (a), what lower bound can you put on $n + (-1)^n$ for $n \geq 2$?

2. Which convergent p-series controls part (a)?

3. In part (b), what is the Nth partial sum after telescoping?

4. Why does that partial sum sequence diverge to $+\infty$?

5. In part (c), what necessary condition for convergence fails?

Part (a).
For $n \geq 2$, show $n + (-1)^n \geq n - 1$, then compare:

Use the comparison test with $\sum 1/(n-1)^2$:

Part (b).
Let $s_n = \sum_{k=1}^{n}(\sqrt{k+1} - \sqrt{k})$. Prove by induction/telescoping that $s_n =$

Show $s_n \to +\infty$, so the series diverges:

Part (c).
Suppose the series converged. Then its terms would satisfy:

Explain why $(-1)^n$ does not satisfy this:

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

Official Solution (Professor)

(a) We will prove that this series converges. For $n \geq 2$, we have

$$\frac{1}{[n+(-1)^n]^2} \leq \frac{1}{(n-1)^2}.$$

From lecture, we know that the series

$$\sum_{n=2}^{\infty} \frac{1}{(n-1)^2} = \sum_{k=1}^{\infty} \frac{1}{k^2}$$

converges. Therefore our series $\sum \frac{1}{[n+(-1)^n]^2}$ also converges by the comparison test.
(b) We will prove that this series diverges. Let

$$s_n = \sum_{k=1}^{n} (\sqrt{k+1} - \sqrt{k})$$

denote the nth partial sum.
Claim: $s_n = \sqrt{n+1} - 1$ for all $n \geq 1$. We will argue using induction.

- Base case: For $n = 1$, we have

$$s_1 = \sqrt{2} - \sqrt{1} = \sqrt{2} - 1.$$

- Inductive step: Suppose that $s_n = \sqrt{n+1} - 1$ for some $n \geq 1$. Then the formula for s_{n+1} holds, since

$$s_{n+1} = s_n + (\sqrt{n+2} - \sqrt{n+1}) = (\sqrt{n+1} - 1) + (\sqrt{n+2} - \sqrt{n+1}) = \sqrt{n+2} - 1.$$

By induction, we conclude that $s_n = \sqrt{n+1} - 1$ holds for all $n \geq 1$.
Notice that the sequence $s_n = \sqrt{n+1} - 1$ diverges to $+\infty$. Indeed, given $M > 0$, pick $N \in \mathbb{N}$ so that $N > (M+1)^2$, and then for any $n \geq N$ we have

$$\sqrt{n+1} - 1 \geq \sqrt{N} - 1 > \sqrt{(M+1)^2} - 1 = M.$$

As the sequence $\{s_n\}$ diverges, then the series $\sum(\sqrt{n+1} - \sqrt{n})$ diverges.
(c) We will prove that this series diverges. Suppose towards a contradiction that $\sum(-1)^n$ converges. Then $\lim(-1)^n = 0$ by Lecture 16. However, we know that the sequence $\{(-1)^n\}$ diverges by Lecture 10, and so this is a contradiction.

Reflection & Variation **Homework 6, Problem 6**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 7

Problem 1

Problem Statement **Homework 7, Problem 1**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \sum_{n=2}^{\infty} \frac{1}{\log n} \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{n-1}{n^2} \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{n^2}{n!}$$

Pass 1 – First Diagnosis **Homework 7, Problem 1**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 7, Problem 1**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 7, Problem 1**

1. Which convergence test is adapted to $\sum 1/\log n$, and why?

2. In part (a), why do the dyadic terms fail the term test?

3. In part (b), how does $(n-1)/n^2$ compare to $1/(2n)$?

4. In part (c), why is the ratio test natural for $n^2/n!$?

5. Which parts converge and which diverge?

Pass 4 – Skeleton Proof **Homework 7, Problem 1**

Part (a).
Reindex with $a_k = 1/\log(k+1)$ and state the dyadic criterion:

Estimate $2^n a_{2^n}$ from below and explain why the terms do not go to zero:

Part (b).
For $n \geq 2$, prove $\frac{n-1}{n^2} \geq \frac{1}{2n}$:

Use comparison with the harmonic series:

Part (c).
Set $a_n = \frac{n^2}{n!}$. Compute a_{n+1}/a_n:

Apply the ratio test:

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

Official Solution (Professor)

(a) We'll use the dyadic criterion. Consider the series $\sum_{k=1}^{\infty} a_k$ with $a_k = \frac{1}{\log(k+1)}$ for $k \geq 1$. (Strictly speaking, in order to apply the dyadic criterion we need a series that starts at $n = 1$, so we reindexed.)
Then $\{a_k\}_{k\geq 1}$ is a decreasing sequence of nonnegative numbers. Therefore, by the dyadic criterion, $\sum_{k=1}^{\infty} a_k$ converges if and only if the series $\sum_{n=0}^{\infty} 2^n a_{2^n}$ converges. Note that

$$2^n a_{2^n} = \frac{2^n}{\log(2^n + 1)} \geq \frac{2^n}{\log(2^{n+1})} = \frac{1}{\log 2} \cdot \frac{2^n}{n+1}.$$

Now, the sequence $\frac{2^n}{n+1}$ diverges to $+\infty$, and so it cannot converge to 0. Multiplying this by the constant $\frac{1}{\log 2}$, we see that $2^n a_{2^n}$ also cannot converge to 0. Therefore the series $\sum 2^n a_{2^n}$ diverges, and thus $\sum a_k$ diverges as well.
(b) We'll use the comparison test. For $n \geq 2$, we have

$$n - 1 \geq n - \frac{n}{2} = \frac{n}{2} \qquad \Longrightarrow \qquad \frac{n-1}{n^2} \geq \frac{n/2}{n^2} = \frac{1}{2n}.$$

From lecture we know that the series $\sum \frac{1}{n}$ diverges, and thus $\sum_{n=2}^{\infty} \frac{1}{2n}$ diverges as well. Therefore, by the comparison test, $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$ also diverges. Note that $\frac{n-1}{n^2}$ is equal to zero for $n = 1$, so $\sum_{n=2}^{\infty} \frac{n-1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{n-1}{n^2}$ are the same series.
(c) We'll use the ratio test. Set $a_n = \frac{n^2}{n!}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^2 \cdot n!}{n^2 \cdot (n+1)!} = \frac{n+1}{n^2} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Therefore

$$\limsup \left| \frac{a_{n+1}}{a_n} \right| = \lim \frac{a_{n+1}}{a_n} = 0 < 1,$$

and so the series converges by the ratio test.

Reflection & Variation **Homework 7, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 7, Problem 2**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)} \sum_{n=1}^{\infty} \frac{n^4}{4^n} \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{n!}{n^4+3} \qquad \text{(c)} \sum_{n=1}^{\infty} \frac{2^n}{n!}$$

Pass 1 – First Diagnosis **Homework 7, Problem 2**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 7, Problem 2**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 7, Problem 2**

1. Why does a polynomial-over-exponential series suggest the ratio test?

---

2. In part (b), what necessary condition for convergence should be checked?

---

3. How does factorial growth compare to $n^4 + 3$?

---

4. In part (c), what ratio do successive terms have?

---

5. Which of these three series fail because the terms do not go to 0?

Pass 4 – Skeleton Proof **Homework 7, Problem 2**

Part (a).
Set $a_n = \frac{n^4}{4^n}$. Compute a_{n+1}/a_n and its limit:

Apply the ratio test:

Part (b).
Show $\frac{n^4+3}{n!} \to 0$, for example using $n! \geq n(n-1)(n-2)(n-3)(n-4)$:

Use HW5#1 to conclude $\frac{n!}{n^4+3} \to +\infty$, hence the series diverges:

Part (c).
Set $a_n = \frac{2^n}{n!}$. Compute a_{n+1}/a_n and apply the ratio test:

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

Official Solution (Professor)

(a) We'll use the ratio test. Set $a_n = \frac{n^4}{4^n}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)^4 \cdot 4^n}{n^4 \cdot 4^{n+1}} = \frac{(n+1)^4}{4n^4} \longrightarrow \frac{1}{4} \qquad \text{as } n \to \infty.$$

Therefore $\limsup \left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{4} < 1$, and so the series converges by the ratio test.

(b) This series diverges. In brief, the sequence $\frac{n!}{n^4+3}$ diverges to $+\infty$, and so it cannot converge to zero. More concretely, notice that for $n \geq 5$ we have

$$n! \geq n(n-1)(n-2)(n-3)(n-4),$$

and so

$$\frac{n^4+3}{n!} \leq \frac{n^4+3}{n(n-1)(n-2)(n-3)(n-4)} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Therefore the sequence $\frac{n!}{n^4+3} \to +\infty$ by HW5#1. In particular, there must exist an N so that $\frac{n!}{n^4+3} \geq 1$ for all $n \geq N$, and so $\frac{n!}{n^4+3}$ cannot converge to 0.

(c) We'll use the ratio test. Set $a_n = \frac{2^n}{n!}$. Then

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1} \cdot n!}{(n+1)! \cdot 2^n} = \frac{2}{n+1} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Therefore $\limsup \left|\frac{a_{n+1}}{a_n}\right| = 0 < 1$, and so the series converges by the ratio test.

Reflection & Variation | **Homework 7, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 7, Problem 3**

For each of the following series, determine if it converges and prove your answer.

$$\text{(a)}\ \sum_{n=1}^{\infty} \frac{1}{n^n} \qquad \text{(b)}\ \sum_{n=1}^{\infty} \frac{n!}{n^n} \qquad \text{(c)}\ \sum_{n=1}^{\infty} \frac{(-1)^n n!}{2^n}$$

Pass 1 – First Diagnosis **Homework 7, Problem 3**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 7, Problem 3

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 7, Problem 3**

1. Why does $\frac{1}{n^n}$ suggest the root test?

---

2. What lecture result gives $(n!)^{1/n}/n \to 1/e$?

---

3. Why does $1/e < 1$ prove convergence in part (b)?

---

4. In part (c), why should you check whether the terms go to 0?

---

5. How does HW5#1 convert $2^n/n! \to 0$ into $n!/2^n \to +\infty$?

Part (a).
Set $a_n = 1/n^n$. Compute $|a_n|^{1/n}$:

Apply the root test:

Part (b).
Set $a_n = n!/n^n$. Compute $|a_n|^{1/n}$:

Use Lecture 15 to identify the limit and apply the root test:

Part (c).
Use Problem 2(c) to know $\frac{2^n}{n!} \to 0$:

Apply HW5#1 to show $\frac{n!}{2^n} \to +\infty$, then conclude the terms do not go to 0:

Use this page for parts (a) and (b).

Use this page for part (c) and final conclusions.

Official Solution (Professor)

(a) We'll use the root test. Set $a_n = \frac{1}{n^n}$. Then

$$|a_n|^{1/n} = \frac{1}{n} \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Therefore $\limsup |a_n|^{1/n} = 0 < 1$, and so the series converges by the root test.
(b) We'll use the root test. Set $a_n = \frac{n!}{n^n}$, and consider the sequence

$$|a_n|^{1/n} = \left(\frac{n!}{n^n}\right)^{1/n} = \frac{(n!)^{1/n}}{n}.$$

In Lecture 15 we showed that this sequence converges to $1/e$. Therefore

$$\limsup |a_n|^{1/n} = 1/e < 1,$$

and so the series converges by the root test.
(c) This series diverges, because the sequence $\frac{(-1)^n n!}{2^n}$ does not converge to 0. Specifically, we know that the sequence $\frac{2^n}{n!}$ converges to 0. (Indeed, we proved in problem 2c that the series $\sum \frac{2^n}{n!}$ converges, and so its terms must converge to 0.) So, by HW5#1, the sequence $\frac{n!}{2^n}$ diverges to $+\infty$. In particular, there exists an N so that $\frac{n!}{2^n} \geq 1$ for all $n \geq N$. Therefore $\left|\frac{(-1)^n n!}{2^n}\right| \geq 1$ for all $n \geq N$, and so the terms $\frac{(-1)^n n!}{2^n}$ in our series cannot converge to 0.

Reflection & Variation **Homework 7, Problem 3**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 7, Problem 4**

Consider the series

$$\sum_{n=1}^{\infty} 2^{(-1)^n - n} = \frac{1}{4} + \frac{1}{2} + \frac{1}{16} + \frac{1}{8} + \frac{1}{64} + \frac{1}{32} + \cdots .$$

(a) Show that the ratio test is inconclusive.

(b) Prove that the series converges using the root test.

Pass 1 – First Diagnosis **Homework 7, Problem 4**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 7, Problem 4

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 7, Problem 4**

1. What are the first few terms of a_{n+1}/a_n?

---

2. Why does the ratio sequence have two subsequential limits?

---

3. What does it mean for the ratio test to be inconclusive?

---

4. Why does the root test handle $2^{(-1)^n-n}$ better than the ratio test?

---

5. What is $\lim |a_n|^{1/n}$?

Pass 4 – Skeleton Proof **Homework 7, Problem 4**

Part (a).
Set $a_n = 2^{(-1)^n - n}$. Compute the first few ratios $|a_{n+1}/a_n|$:

Identify the liminf and limsup of the ratio sequence:

State why this makes the ratio test inconclusive:

Part (b).
Compute $|a_n|^{1/n} = 2^{((-1)^n/n)-1}$:

Use $(-1)^n/n \to 0$ to find the root-test limit:

Conclude convergence by the root test:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Set $a_n = 2^{(-1)^n - n}$. Then the sequence $\left|\frac{a_{n+1}}{a_n}\right|$ is

$$\frac{1/2}{1/4} = 2, \qquad \frac{1/16}{1/2} = \frac{1}{8}, \qquad \frac{1/8}{1/16} = 2, \qquad \frac{1/64}{1/8} = \frac{1}{8}, \ldots$$

In other words, the even terms of this sequence are $\frac{1}{8}$ and the odd terms are 2. Therefore

$$\liminf \left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{8}, \qquad \limsup \left|\frac{a_{n+1}}{a_n}\right| = 2,$$

and so

$$\liminf \left|\frac{a_{n+1}}{a_n}\right| \leq 1 \leq \limsup \left|\frac{a_{n+1}}{a_n}\right|.$$

(b) Set $a_n = 2^{(-1)^n - n}$. Then

$$|a_n|^{1/n} = 2^{\frac{(-1)^n}{n} - 1}.$$

Since $\frac{(-1)^n}{n} \to 0$, we get

$$\lim_{n\to\infty} |a_n|^{1/n} = \frac{1}{2} < 1.$$

Therefore the series converges by the root test.

Reflection & Variation **Homework 7, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement **Homework 7, Problem 5**

Below are two series $\sum_{n=1}^{\infty} a_n$ and the numbers $s \in \mathbb{R}$ to which they converge. For each of these series, what is the set of all numbers $\widetilde{s} \in \mathbb{R}$ such that there exists a rearrangement that satisfies $\sum_{n=1}^{\infty} \widetilde{a}_n = \widetilde{s}$? Prove your answer.

$$\text{(a)} \ \sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\log 2 \qquad \text{(b)} \ \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = -\frac{\pi^2}{12}$$

(Hint: Your proof should be quite short, because we already did most of the work in lecture.)

Pass 1 – First Diagnosis **Homework 7, Problem 5**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 7, Problem 5

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 7, Problem 5**

1. What is the difference between conditional and absolute convergence for rearrangements?

---

2. Why is the alternating harmonic series conditionally convergent?

---

3. What does Riemann's rearrangement theorem say about possible sums?

---

4. Why does $\sum(-1)^n/n^2$ converge absolutely?

---

5. What theorem says every rearrangement of an absolutely convergent series has the same sum?

Pass 4 – Skeleton Proof **Homework 7, Problem 5**

Part (a).
Show $\sum(-1)^n/n$ converges but not absolutely:

Apply Riemann's rearrangement result with $\alpha = \widetilde{s} = \beta$:

Conclude the set of possible rearranged sums is:

Part (b).
Show $\sum(-1)^n/n^2$ converges absolutely:

Apply the theorem on rearrangements of absolutely convergent series:

Conclude the set of possible rearranged sums is:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) We claim that the set is $\mathbb{R}$. Note that:

- $\sum \frac{(-1)^n}{n}$ converges: by the alternating series test, since $\{\frac{1}{n}\}$ is a decreasing sequence that converges to 0.
- $\sum \frac{(-1)^n}{n}$ does not converge absolutely: since $\left|\frac{(-1)^n}{n}\right| = \frac{1}{n}$, and we know $\sum \frac{1}{n}$ diverges by Lecture 11.

Fix $\widetilde{s} \in \mathbb{R}$. Applying the Riemann rearrangement result (with $\alpha = \widetilde{s} = \beta$), we deduce that there exists a rearrangement $\sum \widetilde{a}_n$ that converges to $\widetilde{s}$. This works for any $\widetilde{s} \in \mathbb{R}$, and so $\mathbb{R}$ is the desired set.

(b) We claim that the set is $\{-\frac{\pi^2}{12}\}$. Note that $\sum \frac{(-1)^n}{n^2}$ converges absolutely, since $\left|\frac{(-1)^n}{n^2}\right| = \frac{1}{n^2}$ and we know $\sum \frac{1}{n^2}$ converges by Lecture 16. Suppose that $\sum \widetilde{a}_n$ is a rearrangement that converges to some $\widetilde{s} \in \mathbb{R}$. Applying the second theorem from Lecture 20, we deduce that $\widetilde{s} = -\frac{\pi^2}{12}$. This works for any rearrangement, so $-\frac{\pi^2}{12}$ is the only value that can be attained.

Reflection & Variation **Homework 7, Problem 5**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement **Homework 7, Problem 6**

Prove or disprove the following statements:

(a) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_{2n}$ converges.

(b) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges absolutely.

Pass 1 – First Diagnosis **Homework 7, Problem 6**

Topic: ________________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 7, Problem 6

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 7, Problem 6**

1. Why can ordinary convergence depend on cancellation between odd and even terms?

---

2. What standard alternating example disproves part (a)?

---

3. What is a_{2n} for that example?

---

4. Why does absolute convergence survive passing to a subseries?

---

5. How does the Cauchy criterion prove part (b)?

Pass 4 – Skeleton Proof **Homework 7, Problem 6**

Part (a).
Choose $a_n =$ so that $\sum a_n$ is alternating harmonic:

Compute a_{2n} and identify the divergent series:

State the counterexample conclusion:

Part (b).
Assume $\sum |a_n|$ converges. Use the Cauchy criterion to choose N:

For $n \geq m \geq N$, compare $\sum_{k=m}^{n} |a_{2k}|$ to a tail of $\sum |a_j|$:

Conclude $\sum |a_{2n}|$ converges:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Disprove: Consider the series $a_n = \frac{(-1)^n}{n}$. Then:

- $\sum a_n$ converges: by the alternating series test, since $\frac{1}{n}$ is a decreasing sequence that converges to 0.
- $\sum a_{2n} = \sum \frac{1}{2n}$ diverges: This is $\frac{1}{2}$ times the series $\sum \frac{1}{n}$, which we know diverges by lecture.

(b) Prove: We will use the Cauchy criterion. Fix $\varepsilon > 0$. As $\sum |a_n|$ converges, then there exists an N so that

$$\sum_{k=m}^{n} |a_k| < \varepsilon \qquad \text{for all } n \geq m \geq N.$$

Then for $n \geq m \geq N$, we have

$$\sum_{k=m}^{n} |a_{2k}| \leq \sum_{j=2m}^{2n} |a_j| < \varepsilon.$$

(Note that first sum above consists of only even terms $|a_{2m}| + |a_{2m+2}| + \cdots + |a_{2n}|$, which differs from the second sum $|a_{2m}| + |a_{2m+1}| + |a_{2m+2}| + \cdots + |a_{2n}|$ by only the odd terms, each of which is nonnegative.) Therefore $\sum |a_{2n}|$ converges by the Cauchy criterion.

Reflection & Variation **Homework 7, Problem 6**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 8

Problem 1

Problem Statement **Homework 8, Problem 1**

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its interior and prove your answer.

(a) $A = [0, 1)$

(b) $A = \mathbb{Z}$

(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Pass 1 – First Diagnosis **Homework 8, Problem 1**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 8, Problem 1

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 8, Problem 1**

1. What does it mean for x to be an interior point of A?

---

2. Why is every point of $(0, 1)$ interior to $[0, 1)$?

---

3. Why is 0 not an interior point of $[0, 1)$?

---

4. Why can no open interval in $\mathbb{R}$ be contained in $\mathbb{Z}$?

---

5. How does density of irrational numbers show $\{x \in \mathbb{Q} : 0 < x < 1\}^\circ = \varnothing$?

Part (a).
Show $(0,1) \subseteq A^\circ$ by choosing $r = \frac{1}{2}\min\{x, 1-x\}$:

Show $0 \notin A^\circ$ by finding a point in $(-r, r) \setminus [0,1)$:

Conclude $A^\circ = (0,1)$:

Part (b).
Fix $n \in \mathbb{Z}$ and $r > 0$. Find a noninteger point in $(n-r, n+r)$:

Conclude $\mathbb{Z}^\circ = \varnothing$:

Part (c).

Fix $a \in A$ and $r > 0$. Use density of $\mathbb{R} \setminus \mathbb{Q}$ to find $y \in (a - r, a + r) \setminus A$:

Pass 5 – Full Proof Attempt Homework 8, Problem 1

Use this page for parts (a) and (b).

Pass 5 – Full Proof Attempt Homework 8, Problem 1

Use this page for part (c) and final conclusions.

Official Solution (Professor)

(a) Claim: $A^\circ = (0,1)$. As A° is the largest open subset of A, and $(0,1)$ is one open subset, we must have $(0,1) \subseteq A^\circ \subseteq [0,1)$. Note that 0 is not an interior point: given $r > 0$ we have $(-r,r) \nsubseteq [0,1)$, since $-\frac{r}{2}$ is a point in $(-r,r)$ that is not in $[0,1)$.
(b) Claim: $A^\circ = \varnothing$. Fix $n \in \mathbb{Z}$; we will show that n is not an interior point. Given $r > 0$ we have $(n-r, n+r) \nsubseteq \mathbb{Z}$, since $\min\{n + \frac{r}{2}, n + \frac{1}{2}\}$ is in $(n-r, n+r)$ but not in $\mathbb{Z}$.
(c) Claim: $A^\circ = \varnothing$. Fix $a \in A$; we will show that a is not an interior point. As the irrational numbers are dense in $\mathbb{R}$, then for any $r > 0$ we can find a point $y \in \mathbb{Q}^c$ between $a - r$ and a. This y is a point in $(a-r, a+r)$ that is not in A, and so $(a-r, a+r) \nsubseteq A$.

Reflection & Variation **Homework 8, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 8, Problem 2**

Recall that d_1 and d_∞ are both metrics on $\mathbb{R}^2$.

(a) Prove that $d_\infty(x,y) \le d_1(x,y) \le 2d_\infty(x,y)$ for all $x, y \in \mathbb{R}^2$.

(b) Prove that for any $x \in \mathbb{R}^2$ and $r > 0$ we have

$$\{y \in \mathbb{R}^2 : d_\infty(x,y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x,y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x,y) < r\}.$$

I recommend that you draw a picture of these three sets.

(c) Prove that a set $A \subseteq \mathbb{R}^2$ is open in the metric space $(\mathbb{R}^2, d_1)$ if and only if it is open in the metric space $(\mathbb{R}^2, d_\infty)$.

Pass 1 – First Diagnosis **Homework 8, Problem 2**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 8, Problem 2

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 8, Problem 2**

1. What are the formulas for $d_1(x, y)$ and $d_\infty(x, y)$ in coordinates?

2. Why is the maximum of two nonnegative numbers at most their sum?

3. Why is the sum of two nonnegative numbers at most twice their maximum?

4. How do metric inequalities translate into ball containments?

5. How do the ball containments prove the two metrics define the same open sets?

Pass 4 – Skeleton Proof **Homework 8, Problem 2**

Part (a).
Write $x = (x_1, x_2)$, $y = (y_1, y_2)$, and expand d_1, d_∞:

Prove $d_\infty \leq d_1$:

Prove $d_1 \leq 2d_\infty$:

Part (b).
If $d_\infty(x, y) < r/2$, use part (a) to prove $d_1(x, y) < r$:

If $d_1(x, y) < r$, use part (a) to prove $d_\infty(x, y) < r$:

Part (c).

For d_∞-open $\Rightarrow$ d_1-open, use $B_r^1(a) \subseteq B_r^\infty(a)$:

For d_1-open $\Rightarrow$ d_∞-open, use $B_{r/2}^\infty(a) \subseteq B_r^1(a)$:

Use this page for parts (a) and (b).

Use this page for part (c).

Official Solution (Professor)

(a) Fix $x = (x_1, x_2)$ and $y = (y_1, y_2)$ in $\mathbb{R}^2$. Then

$$|x_1 - y_1| \leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y),$$

$$|x_2 - y_2| \leq |x_1 - y_1| + |x_2 - y_2| = d_1(x, y),$$

and so

$$\max\{|x_1 - y_1|, |x_2 - y_2|\} \leq d_1(x, y).$$

This proves the first inequality: $d_\infty(x, y) \leq d_1(x, y)$. On the other hand, note that

$$|x_1 - y_1| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y),$$

$$|x_2 - y_2| \leq \max\{|x_1 - y_1|, |x_2 - y_2|\} = d_\infty(x, y),$$

and so

$$|x_1 - y_1| + |x_2 - y_2| \leq 2d_\infty(x, y).$$

This proves that $d_1(x, y) \leq 2d_\infty(x, y)$.
(b) Let's start with the first containment:

$$\{y \in \mathbb{R}^2 : d_\infty(x, y) < r/2\} \subseteq \{y \in \mathbb{R}^2 : d_1(x, y) < r\}.$$

Fix $y \in \mathbb{R}^2$ so that $d_\infty(x, y) < r/2$. Then by the second inequality in part (a) we have

$$d_1(x, y) \leq 2d_\infty(x, y) < 2 \cdot \frac{r}{2} = r,$$

and so $d_1(x, y) < r$.
Next, we turn to the second containment:

$$\{y \in \mathbb{R}^2 : d_1(x, y) < r\} \subseteq \{y \in \mathbb{R}^2 : d_\infty(x, y) < r\}.$$

Fix $y \in \mathbb{R}^2$ so that $d_1(x, y) < r$. Then by the first inequality in part (a), we have

$$d_\infty(x, y) \leq d_1(x, y) < r,$$

and so $d_\infty(x, y) < r$.
(c) $\Leftarrow$ Suppose that A is open in $(\mathbb{R}^2, d_\infty)$. Fix $a \in A$. Then there exists $r > 0$ so that

$$B_r^\infty(a) = \{x \in \mathbb{R}^2 : d_\infty(x, a) < r\} \subseteq A.$$

Note that $B_r^1(a) \subseteq B_r^\infty(a)$ by part (b), and so $B_r^1(a) \subseteq B_r^\infty(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(\mathbb{R}^2, d_1)$.
$\Rightarrow$ Suppose that A is open in $(\mathbb{R}^2, d_1)$. Fix $a \in A$. Then there exists $r > 0$ so that $B_r^1(a) \subseteq A$. Note that $B_{r/2}^\infty(a) \subseteq B_r^1(a)$ by part (b), and so $B_{r/2}^\infty(a) \subseteq B_r^1(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(\mathbb{R}^2, d_\infty)$.

Reflection & Variation | **Homework 8, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 8, Problem 3**

Let X be a nonempty set, and recall that the discrete metric d is a metric on X. Prove that every set $A \subseteq X$ is open in the metric space (X, d).

Pass 1 – First Diagnosis **Homework 8, Problem 3**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 8, Problem 3

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 8, Problem 3**

1. What is the definition of the discrete metric?

---

2. What is $B_{1/2}(a)$ in the discrete metric?

---

3. Why is $\{a\} \subseteq A$ whenever $a \in A$?

---

4. How does this prove every point of A is interior?

---

5. Why does the proof also work when $A = \varnothing$?

Pass 4 – Skeleton Proof **Homework 8, Problem 3**

Fix $A \subseteq X$ and $a \in A$:

Compute $B_{1/2}(a) = \{x \in X : d(x, a) < 1/2\}$:

Use $B_{1/2}(a) = \{a\} \subseteq A$ to show a is interior:

Conclude A is open because $a \in A$ was arbitrary:

Use this page for the complete proof.

Official Solution (Professor)

Fix $A \subseteq X$ and $a \in A$. Notice that

$$B_{1/2}(a) = \{x \in X : d(x, a) < 1/2\} = \{a\}.$$

(Indeed, $d(x, a)$ is either equal to 0 or 1, and so the only time that $d(x, a) < 1/2$ is when $d(x, a) = 0$.) Then $a \in B_{1/2}(a) \subseteq A$, and so a is an interior point for A. As $a \in A$ was arbitrary, we conclude that A is open.

Reflection & Variation | Homework 8, Problem 3

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 8, Problem 4**

Let (X, d) be any metric space. Define the function $\widetilde{d} : X \times X \to \mathbb{R}$ by

$$\widetilde{d}(x, y) = \min\{d(x, y), 1\}.$$

(a) Prove that $\widetilde{d}$ is also a metric on X.

(b) Prove that a set $A \subseteq X$ is open in (X, d) if and only if it is open in $(X, \widetilde{d})$.

Pass 1 – First Diagnosis **Homework 8, Problem 4**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 8, Problem 4**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 8, Problem 4**

1. Which metric axioms are immediate from the definition of $\widetilde{d}$?

---

2. Why is definiteness preserved by taking $\min\{d(x, y), 1\}$?

---

3. What are the key cases for the triangle inequality?

---

4. Why do d and $\widetilde{d}$ agree on distances less than 1?

---

5. Why does shrinking a radius to $\min\{r, 1/2\}$ transfer openness?

Pass 4 – Skeleton Proof **Homework 8, Problem 4**

Part (a).
Check nonnegativity, definiteness, and symmetry:

For the triangle inequality, split into cases depending on whether $d(x, y) < 1$ and $d(y, z) < 1$:

Conclude $\widetilde{d}$ is a metric:

Part (b).
Assume A is open in $(X, \widetilde{d})$. Use $\widetilde{d}(x, a) \leq d(x, a)$ to get a d-ball inside A:

Assume A is open in (X, d). Set $r_2 = \min\{r_1, 1/2\}$ and prove the $\widetilde{d}$-ball is inside the d-ball:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) There are 4 properties to check.

1. $\widetilde{d}(x,y) \geq 0$ for all $x,y \in X$. Note that $\widetilde{d}(x,y)$ is equal to 1 or $d(x,y)$. Both of these are nonnegative; we know $d(x,y) \geq 0$ since d is a metric on X.
2. $\widetilde{d}(x,y) = 0 \iff x = y$. If $x = y$, then $d(x,y) = 0$, so $\widetilde{d}(x,y) = \min\{0,1\} = 0$. Conversely, if $\widetilde{d}(x,y) = 0$, then $\widetilde{d}(x,y) \neq 1$ and so we must have $d(x,y) = \widetilde{d}(x,y) = 0$. As d is a metric, this can only mean that $x = y$.
3. $\widetilde{d}(x,y) = \widetilde{d}(y,x)$ for all $x,y \in X$. As d is a metric, we know that $d(x,y) = d(y,x)$, and so
$$\widetilde{d}(x,y) = \min\{d(x,y),1\} = \min\{d(y,x),1\} = \widetilde{d}(y,x).$$
4. $\widetilde{d}(x,z) \leq \widetilde{d}(x,y) + \widetilde{d}(y,z)$ for all $x,y,z \in X$. We distinguish 4 possible cases:
   - $d(x,y) < 1$ and $d(y,z) < 1$: Then
   $$\widetilde{d}(x,z) \leq d(x,z) \leq d(x,y) + d(y,z) = \widetilde{d}(x,y) + \widetilde{d}(y,z).$$
   - $d(x,y) < 1$ and $d(y,z) \geq 1$: Then
   $$\widetilde{d}(x,z) \leq 1 \leq d(x,y) + 1 = \widetilde{d}(x,y) + \widetilde{d}(y,z).$$
   - $d(x,y) \geq 1$ and $d(y,z) < 1$: Then
   $$\widetilde{d}(x,z) \leq 1 \leq 1 + d(y,z) = \widetilde{d}(x,y) + \widetilde{d}(y,z).$$
   - $d(x,y) \geq 1$ and $d(y,z) \geq 1$: Then
   $$\widetilde{d}(x,z) \leq 1 \leq 1 + 1 = \widetilde{d}(x,y) + \widetilde{d}(y,z).$$

In all cases, the triangle inequality holds.

(b) $\Leftarrow$ Suppose that A is open in $(X,\widetilde{d})$. Fix $a \in A$. Then there exists $r > 0$ so that
$$B_r^{\widetilde{d}}(a) = \{x \in X : \widetilde{d}(x,a) < r\} \subseteq A.$$
Note that $B_r^d(a) \subseteq B_r^{\widetilde{d}}(a)$, since
$$x \in B_r^d(a) \Longrightarrow \widetilde{d}(x,a) \leq d(x,a) < r \Longrightarrow x \in B_r^{\widetilde{d}}(a).$$
So $B_r^d(a) \subseteq B_r^{\widetilde{d}}(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in (X,d).

$\Rightarrow$ Suppose that A is open in (X,d). Fix $a \in A$. Then there exists $r_1 > 0$ so that $B_{r_1}^d(a) \subseteq A$. Set $r_2 = \min\{r_1, \frac{1}{2}\}$. Note that $B_{r_2}^{\widetilde{d}}(a) \subseteq B_{r_1}^d(a)$, since
$$x \in B_{r_2}^{\widetilde{d}}(a) \Longrightarrow \widetilde{d}(x,a) = \min\{d(x,a),1\} < r_2 \leq \frac{1}{2}$$
$$\Longrightarrow \widetilde{d}(x,a) = d(x,a) \Longrightarrow d(x,a) < r_2 \leq r_1.$$
So $B_{r_2}^{\widetilde{d}}(a) \subseteq B_{r_1}^d(a) \subseteq A$. As $a \in A$ was arbitrary, we conclude that A is open in $(X,\widetilde{d})$.

Reflection & Variation **Homework 8, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement **Homework 8, Problem 5**

Let (X, d) be a metric space, $A \subseteq X$, and $x \in X$. Prove that x is in A° if and only if there exists an open set U so that $x \in U \subseteq A$.

Pass 1 – First Diagnosis **Homework 8, Problem 5**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 8, Problem 5

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 8, Problem 5**

1. What is the definition of $x \in A^\circ$?

---

2. Why is an open ball an open set?

---

3. In the forward direction, what should U be?

---

4. In the reverse direction, why does openness of U give a ball around x?

---

5. How does $B_r(x) \subseteq U \subseteq A$ finish the proof?

Pass 4 – Skeleton Proof **Homework 8, Problem 5**

Forward direction.
Assume $x \in A^\circ$. Choose $r > 0$ with $B_r(x) \subseteq A$:

Set $U =$ and verify open, $x \in U$, and $U \subseteq A$:

Reverse direction.
Assume $x \in U \subseteq A$ with U open. Choose $r > 0$ with:

Use $B_r(x) \subseteq U \subseteq A$ to conclude $x \in A^\circ$:

Use this page for both directions.

Official Solution (Professor)

$\Rightarrow$ Suppose that $x \in A^\circ$. Then there exists $r > 0$ such that $B_r(x) \subseteq A$. Notice that $U = B_r(x)$ is an open set that satisfies $x \in U \subseteq A$.
$\Leftarrow$ Suppose that $x \in U \subseteq A$ for some open set U. As U is open, there exists $r > 0$ so that $B_r(x) \subseteq U$. Together we have $B_r(x) \subseteq U \subseteq A$, and so $x \in A^\circ$.

Reflection & Variation **Homework 8, Problem 5**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement **Homework 8, Problem 6**

Let (X, d) be a metric space and let A be a nonempty subset of X. Prove that A is open if and only if there exists a set I, positive numbers $r_i > 0$, and points $a_i \in X$ so that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Pass 1 – First Diagnosis | Homework 8, Problem 6

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 8, Problem 6**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 8, Problem 6**

1. Why should open sets be expressible as unions of balls?

2. In the forward direction, what index set I is the natural choice?

3. Why does every point $a \in A$ belong to its chosen ball $B_{r_a}(a)$?

4. Why is each ball $B_{r_i}(a_i)$ open?

5. Which theorem says arbitrary unions of open sets are open?

Pass 4 – Skeleton Proof **Homework 8, Problem 6**

Forward direction.
Assume A is open. For each $a \in A$, choose $r_a > 0$ with $B_{r_a}(a) \subseteq A$:

Use $I = A$, and for each $i = a \in A$, set the center $a_i = a$ and radius $r_i = r_a$:

Prove both inclusions $A \subseteq \bigcup_{a \in A} B_{r_a}(a)$ and $\bigcup_{a \in A} B_{r_a}(a) \subseteq A$:

Reverse direction.
Assume $A = \bigcup_{i \in I} B_{r_i}(a_i)$:

Each ball is open, and arbitrary unions of open sets are open. Conclude:

Pass 5 – Full Proof Attempt Homework 8, Problem 6

Use this page for the forward direction.

Use this page for the reverse direction.

Student-Derived Solution (No official solution available)

We prove both directions.
Suppose first that A is open. Since A is nonempty, for each $a \in A$ choose $r_a > 0$ such that

$$B_{r_a}(a) \subseteq A.$$

Then

$$A = \bigcup_{a \in A} B_{r_a}(a).$$

Indeed, if $x \in A$, then $x \in B_{r_x}(x)$, so $x \in \bigcup_{a \in A} B_{r_a}(a)$. Conversely, each $B_{r_a}(a) \subseteq A$, so the union is contained in A. Thus A has the required form, with index set $I = A$, and for each index $i = a \in A$, center $a_i = a$ and radius $r_i = r_a$.
Conversely, suppose there exist a set I, positive numbers $r_i > 0$, and points $a_i \in X$ such that

$$A = \bigcup_{i \in I} B_{r_i}(a_i).$$

Each ball $B_{r_i}(a_i)$ is open in (X, d), and an arbitrary union of open sets is open. Therefore A is open.

Reflection & Variation **Homework 8, Problem 6**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 9

Problem 1

Problem Statement **Homework 9, Problem 1**

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. For each of the following sets $A \subseteq \mathbb{R}$, find its closure, isolated points, and limit points. Prove your answer.

(a) $A = [0, 1)$

(b) $A = \mathbb{Z}$

(c) $A = \{x \in \mathbb{Q} : 0 < x < 1\}$

Pass 1 – First Diagnosis **Homework 9, Problem 1**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 9, Problem 1

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 9, Problem 1**

1. For each set, which points are obvious members of the closure, and which boundary points require a ball argument?

---

2. How do you prove that a point is not in $\overline{\mathbb{Z}}$? What radius separates it from all integers?

---

3. What is the difference between proving $a \in A$ is isolated and proving $a \in A$ is a limit point?

---

4. Where is density of $\mathbb{Q}$ used in part (c), and why is it not enough just to say "there are rationals nearby"?

---

5. How does the identity $\overline{A} = (\text{isolated points}) \cup A'$ finish the limit-point computations?

Pass 4 – Skeleton Proof **Homework 9, Problem 1**

Part (a): $A = [0, 1)$
Closure: show $[0, 1) \subseteq \overline{A} \subseteq [0, 1]$**, then prove** $1 \in \overline{A}$ **by choosing a point of** A **in** $(1 - r, 1 + r)$**:**

Isolated points: fix $a \in A$**. Given** $r > 0$**, choose a second point of** A **inside** $(a - r, a + r)$**:**

Limit points: use the closure/isolated-point relationship to conclude:

Part (b): $A = \mathbb{Z}$
Closure: for $x \notin \mathbb{Z}$**, choose** $n \in \mathbb{Z}$ **with** $n < x < n + 1$ **and radius** $r =$

Isolated points and limit points: choose the ball around $n \in \mathbb{Z}$**:**

Pass 4 – Skeleton Proof **Homework 9, Problem 1 (continued)**

Part (c): $A = \{x \in \mathbb{Q} : 0 < x < 1\}$
Closure: prove 0**, points in** $(0, 1]$**, and no points outside** $[0, 1]$ **behave correctly:**

Isolated points and limit points: use density of $\mathbb{Q}$ **to find a second point:**

Pass 5 – Full Proof Attempt Homework 9, Problem 1

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

Official Solution (Professor)

(a) Closure: $\overline{A} = [0,1]$. Since $\overline{A}$ is the smallest closed set containing A, and $[0,1]$ is one such closed set, we must have

$$[0,1) \subseteq \overline{A} \subseteq [0,1].$$

Also $1 \in \overline{A}$: given $r > 0$,

$$(1-r, 1+r) \cap [0,1) \neq \varnothing,$$

since $\max\{1 - r/2, 0\} \in (1-r, 1+r) \cap [0,1)$.
Isolated points: $\varnothing$. Fix $a \in A$. Given $r > 0$, the point

$$\min\left\{a + \frac{r}{2}, \frac{a+1}{2}\right\}$$

is in $(a-r, a+r) \cap A$ and is not a.
Limit points: $A' = [0,1]$. By definition, $\overline{A}$ is the union of isolated points and limit points, and there are no isolated points.

(b) Closure: $\overline{A} = \mathbb{Z}$. We automatically have $\mathbb{Z} \subseteq \overline{A}$. Fix $x \in \mathbb{Z}^c$. Let $n \in \mathbb{Z}$ satisfy $n < x < n+1$. Then

$$r = \min\{x - n, n + 1 - x\}$$

is positive and $(x-r, x+r) \cap \mathbb{Z} = \varnothing$. Thus $x \notin \overline{A}$.
Isolated points: $\mathbb{Z}$. For $n \in \mathbb{Z}$,

$$(n-1, n+1) \cap \mathbb{Z} = \{n\}.$$

Limit points: $A' = \varnothing$. Every point of A is isolated.

(c) Closure: $\overline{A} = [0,1]$. First, $A \subseteq [0,1]$, and $[0,1]$ is closed, so $\overline{A} \subseteq [0,1]$. The point 0 is adherent to A: given $r > 0$, choose $n > \max\{1/r, 1\}$, so $1/n \in (-r, r) \cap A$. Now fix $x \in (0,1]$. By density of $\mathbb{Q}$ in $\mathbb{R}$, for every $r > 0$ there is $q \in \mathbb{Q}$ between $\max\{x - r, 0\}$ and x, and this $q \in (x-r, x+r) \cap A$.
Isolated points: $\varnothing$. Given $x \in A$ and $r > 0$, choose $q \in \mathbb{Q}$ as above. This q lies in $(x-r, x+r) \cap A$ and $q \neq x$.
Limit points: $A' = [0,1]$. There are no isolated points.

Reflection & Variation **Homework 9, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 9, Problem 2**

Let (X, d) be a metric space, let A be a nonempty subset of X, and let U be an open subset of X.

(a) Prove that $U \cap \overline{A} \subseteq \overline{U \cap A}$.

(b) Prove that $\overline{U \cap \overline{A}} = \overline{U \cap A}$. (Hint: Use part (a) and properties of closure from lecture.)

(c) Conclude that if $U \cap A = \varnothing$, then $U \cap \overline{A} = \varnothing$.

Pass 1 – First Diagnosis **Homework 9, Problem 2**

Topic: ____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 9, Problem 2**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 9, Problem 2**

1. Why is the assumption that U is open essential in part (a)?

---

2. After fixing $x \in U \cap \overline{A}$ and $r > 0$, why do we introduce a second radius $s > 0$?

---

3. Why is $\tilde{r} = \min\{r, s\}$ the right radius to use?

---

4. Which closure property turns part (a) into the first inclusion in part (b)?

---

5. In part (c), why does $\overline{U \cap \overline{A}} = \varnothing$ force $U \cap \overline{A} = \varnothing$?

Pass 4 – Skeleton Proof **Homework 9, Problem 2**

Part (a): prove $U \cap \overline{A} \subseteq \overline{U \cap A}$.
Fix $x \in$ ____________________. To show $x \in \overline{U \cap A}$, fix $r > 0$ and aim to find a point in:

Use openness of U: since $x \in U$, choose $s > 0$ such that:

Use $x \in \overline{A}$ with $\tilde{r} = \min\{r, s\}$ to choose $y \in$:

Check the three memberships $y \in A$, $y \in U$, and $y \in B_r(x)$:

Pass 4 – Skeleton Proof **Homework 9, Problem 2 (continued)**

Part (b): prove equality of closures.
First inclusion from part (a) and monotonicity of closure:

Second inclusion from $A \subseteq \overline{A}$:

Part (c): empty-set conclusion.
Start from $U \cap A = \varnothing$, take closures, use part (b), and then use $B \subseteq \overline{B}$:

Pass 5 – Full Proof Attempt Homework 9, Problem 2

Use this page for parts (a) and (b).

Use this page for part (c) and final cleanup.

Official Solution (Professor)

(a) Fix $x \in U \cap \overline{A}$. We show $x \in \overline{U \cap A}$, i.e., for any $r > 0$, $B_r(x) \cap U \cap A$ is nonempty. Fix $r > 0$. Since $x \in U$ and U is open, there exists $s > 0$ such that $B_s(x) \subseteq U$. Set $\tilde{r} = \min\{r, s\}$. Since $x \in \overline{A}$, the set $B_{\tilde{r}}(x) \cap A$ is nonempty, so choose $y \in B_{\tilde{r}}(x) \cap A$.
Then $y \in A$. Also $y \in B_r(x)$, since $d(y,x) < \tilde{r} \leq r$. Finally $y \in U$, since $d(y,x) < \tilde{r} \leq s$, so $y \in B_s(x) \subseteq U$. Hence $y \in B_r(x) \cap U \cap A$, as desired.

(b) Recall that $B \subseteq C$ implies $\overline{B} \subseteq \overline{C}$. Applying this to part (a),

$$\overline{U \cap \overline{A}} \subseteq \overline{U \cap A}.$$

On the other hand, $A \subseteq \overline{A}$, so $U \cap A \subseteq U \cap \overline{A}$. Therefore

$$\overline{U \cap A} \subseteq \overline{U \cap \overline{A}}.$$

Together,

$$\overline{U \cap \overline{A}} = \overline{U \cap A}.$$

(c) For any set B, $B \subseteq \overline{B}$. If $U \cap A = \varnothing$, then $\overline{U \cap A} = \varnothing$. By part (b),

$$\overline{U \cap \overline{A}} = \varnothing.$$

Therefore $U \cap \overline{A} = \varnothing$.

Reflection & Variation **Homework 9, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 9, Problem 3**

Recall that d_2 is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, 0) : x \in \mathbb{R}\}$$

with the induced metric d_Y.

(a) Provide an example of a set $A \subseteq Y$ where A is open in (Y, d_Y) but not open in $(\mathbb{R}^2, d_2)$.

(b) Prove that for any $B \subseteq Y$, if B is closed in (Y, d_Y) then B is closed in $(\mathbb{R}^2, d_2)$.

Pass 1 – First Diagnosis **Homework 9, Problem 3**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 9, Problem 3

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 9, Problem 3**

1. What does a ball in the subspace Y look like compared with a ball in $\mathbb{R}^2$?

---

2. Why is the line segment $(-1, 1) \times \{0\}$ open in Y?

---

3. Why does the point $(0, r/2)$ prove that this set is not open in $\mathbb{R}^2$?

---

4. What lecture result lets us reduce part (b) to proving that Y is closed in $\mathbb{R}^2$?

---

5. How does the vertical coordinate $x_2 \neq 0$ give a ball contained in Y^c?

Pass 4 – Skeleton Proof **Homework 9, Problem 3**

Part (a): example open in Y, not open in $\mathbb{R}^2$.
Choose $A =$

To prove A is open in Y, fix $x = (x_1, 0) \in A$ and choose $r =$

If $y \in B_r^Y(x)$, show $|y_1| < 1$, hence $y \in A$:

To prove A is not open in $\mathbb{R}^2$, fix $r > 0$ around $(0, 0)$ and find an off-axis point:

Part (b): closedness transfer.
State the lecture corollary being used:

Pass 4 – Skeleton Proof **Homework 9, Problem 3 (continued)**

Part (b): closedness transfer, continued.
Prove Y is closed by fixing $x = (x_1, x_2) \in Y^c$, so $x_2 \neq 0$, and choosing radius $r =$

Show $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$ using the inequality involving $|x_2 - y_2|$:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Consider
$$A = \{(x_1, 0) : -1 < x_1 < 1\}.$$
First, A is open in (Y, d_Y). Fix $x = (x_1, 0) \in A$, and set $r = 1 - |x_1|$. Then $r > 0$, and
$$B_r^Y(x) = \{y \in Y : d_Y(x, y) < r\} \subseteq A,$$
since
$$y \in B_r^Y(x) \Longrightarrow r > d_Y(x, y) = \sqrt{|x_1 - y_1|^2 + |0 - 0|^2} = |x_1 - y_1|$$
and therefore
$$|y_1| \leq |x_1| + |y_1 - x_1| < |x_1| + r = 1.$$
Next, A is not open in $(\mathbb{R}^2, d_2)$, because $(0, 0) \in A$ is not an interior point. For any $r > 0$, the point $(0, r/2)$ lies in $B_r^{\mathbb{R}^2}((0, 0))$ but not in A.

(b) By the corollary from Lecture 24, it suffices to show that Y is closed in $(\mathbb{R}^2, d_2)$. Fix $x = (x_1, x_2) \in Y^c$. Then $x_2 \neq 0$. Set $r = |x_2|$. If $y \in B_r^{\mathbb{R}^2}(x)$, then
$$r > d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2} \geq |x_2 - y_2|.$$
Hence
$$|y_2| \geq |x_2| - |x_2 - y_2| > |x_2| - r = 0.$$
Thus $y_2 \neq 0$, so $y \in Y^c$. Therefore $B_r^{\mathbb{R}^2}(x) \subseteq Y^c$. Since every point of Y^c is interior to Y^c, Y^c is open, and Y is closed.

Reflection & Variation **Homework 9, Problem 3**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 9, Problem 4**

Let (X, d) be a metric space, $\{x_n\}_{n\geq 1}$ a sequence in X, and $x \in X$. Prove that the sequence $\{x_n\}_{n\geq 1}$ converges to x if and only if for every open set U that contains x, we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$.

Pass 1 – First Diagnosis Homework 9, Problem 4

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | Homework 9, Problem 4

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 9, Problem 4**

1. In the forward direction, how does openness of U connect an arbitrary neighborhood to a ball around x?

---

2. What does "for all but finitely many n" mean in terms of choosing an N?

---

3. In the reverse direction, why is $B_r(x)$ the correct open set to test?

---

4. Why is it enough to prove $x_n \in B_r(x)$ eventually for each $r > 0$?

---

5. What is the main logical difference between the two directions?

Pass 4 – Skeleton Proof **Homework 9, Problem 4**

Forward direction: assume $x_n \to x$.
Fix an open set $U \subseteq X$ with $x \in U$. By openness, choose $r > 0$ such that:

By convergence, choose N such that for all $n \geq N$:

Conclude $x_n \in U$ for all but finitely many n:

Reverse direction: assume eventual membership in every open $U \ni x$.
Fix $r > 0$. Apply the hypothesis to the open set:

Convert "all but finitely many" into an N so that for all $n \geq N$:

Translate $x_n \in B_r(x)$ into the metric convergence inequality:

Conclude because $r > 0$ was arbitrary:

Pass 5 – Full Proof Attempt **Homework 9, Problem 4**

Use this page for both directions.

Official Solution (Professor)

$\Rightarrow$ Suppose $x_n \to x$ in (X, d) as $n \to \infty$. Fix an open set $U \subseteq X$ such that $x \in U$. Since U is open, there exists $r > 0$ such that $B_r(x) \subseteq U$. By definition of convergence, there exists N such that $d(x_n, x) < r$ for all $n \geq N$. Therefore $x_n \in B_r(x) \subseteq U$ for all $n \geq N$.

$\Leftarrow$ Suppose that for every open set U containing x, we have $x_n \in U$ for all but finitely many $n \in \mathbb{N}$. Fix $r > 0$. Apply the assumption to the open set $B_r(x)$. Then $x_n \in B_r(x)$ for all but finitely many $n \in \mathbb{N}$. Let N be larger than the largest index for which $x_n \notin B_r(x)$. Then for every $n \geq N$, $x_n \in B_r(x)$, and hence $d(x_n, x) < r$. Since $r > 0$ was arbitrary, $\{x_n\}_{n \geq 1}$ converges to x in (X, d).

Reflection & Variation **Homework 9, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement **Homework 9, Problem 5**

Recall that d_2 is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$.

(a) Prove that $|x_1^n - x_1| \leq d_2(x^n, x)$ for any $x = (x_1, x_2) \in \mathbb{R}^2$.

(b) Prove that the sequence $\{x^n\}_{n\geq 1}$ converges in $(\mathbb{R}^2, d_2)$ if and only if the sequences $\{x_1^n\}_{n\geq 1}$ and $\{x_2^n\}_{n\geq 1}$ of real numbers both converge.

Pass 1 – First Diagnosis **Homework 9, Problem 5**

Topic: ______________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 9, Problem 5

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 9, Problem 5**

1. Why does the Euclidean distance $d_2(x^n, x)$ control each coordinate difference?

---

2. In the forward direction, what is squeezed between 0 and $d_2(x^n, x)$?

---

3. Why do the coordinate sequences converge to the coordinates of the same point x?

---

4. In the reverse direction, how do the real limit laws imply $d_2(x^n, x) \to 0$?

---

5. What would change if the problem were in $\mathbb{R}^3$ or $\mathbb{R}^m$?

Pass 4 – Skeleton Proof **Homework 9, Problem 5**

Part (a): coordinate estimate.
Write the formula for $d_2(x^n, x)$:

Use $|x_2^n - x_2|^2 \geq 0$ to get the desired inequality:

Part (b), forward direction.
Assume $x^n \to x = (x_1, x_2)$ in $(\mathbb{R}^2, d_2)$. Then $d_2(x^n, x) \to$

Use part (a) and squeeze to prove $x_1^n \to x_1$:

Repeat the same argument for the second coordinate:

Pass 4 – Skeleton Proof **Homework 9, Problem 5 (continued)**

Part (b), reverse direction.
Assume $x_1^n \to x_1$ and $x_2^n \to x_2$. Set $x =$

Use the formula for $d_2(x^n, x)$ and the limit laws to conclude:

Use this page for part (a) and the forward direction of part (b).

Use this page for the reverse direction of part (b).

Official Solution (Professor)

(a) Recall that

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2}.$$

Since $|x_2^n - x_2|^2 \geq 0$,

$$d_2(x^n, x) \geq \left(|x_1^n - x_1|^2 + 0\right)^{1/2} \geq |x_1^n - x_1|.$$

(b) $\Rightarrow$ Suppose $\{x^n\}_{n\geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$. Then

$$d_2(x^n, x) \to 0.$$

By part (a),

$$0 \leq |x_1^n - x_1| \leq d_2(x^n, x).$$

By the squeeze theorem, $|x_1^n - x_1| \to 0$, so $\{x_1^n\}_{n\geq 1}$ converges to x_1. Similarly,

$$0 \leq |x_2^n - x_2| \leq d_2(x^n, x),$$

so $\{x_2^n\}_{n\geq 1}$ converges to x_2.

$\Leftarrow$ Suppose $\{x_1^n\}_{n\geq 1}$ converges to x_1 and $\{x_2^n\}_{n\geq 1}$ converges to x_2. Then

$$|x_1^n - x_1| \to 0 \qquad \text{and} \qquad |x_2^n - x_2| \to 0.$$

By the limit laws,

$$d_2(x^n, x) = \left(|x_1^n - x_1|^2 + |x_2^n - x_2|^2\right)^{1/2} \to 0.$$

Thus $\{x^n\}_{n\geq 1}$ converges to x in $(\mathbb{R}^2, d_2)$.

Reflection & Variation **Homework 9, Problem 5**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement **Homework 9, Problem 6**

Let (X, d) be a metric space and $\{x_n\}_{n\geq 1}$ a sequence in X. Prove that if $\{x_n\}_{n\geq 1}$ converges to $x \in X$, then for any $y \in X$, the sequence $\{d(x_n, y)\}_{n\geq 1}$ of real numbers converges to $d(x, y)$.

Pass 1 – First Diagnosis **Homework 9, Problem 6**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 9, Problem 6**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 9, Problem 6**

1. What real-valued sequence are we trying to prove converges?

---

2. Why should the triangle inequality compare $d(x_n, y)$ and $d(x, y)$?

---

3. How do two applications of the triangle inequality produce an absolute value estimate?

---

4. What quantity goes to 0, and why does it squeeze the distance error?

---

5. How is this problem a sequence version of continuity of the map $z \mapsto d(z, y)$?

Pass 4 – Skeleton Proof **Homework 9, Problem 6**

Setup.
Fix $y \in X$. Since $x_n \to x$, we know $d(x_n, x) \to$

Triangle inequality estimates.
First apply triangle inequality to bound $d(x_n, y)$ in terms of $d(x_n, x)$ and $d(x, y)$:

Rearrange to get $d(x_n, y) - d(x, y) \leq$

Apply triangle inequality again to bound $d(x, y)$ in terms of $d(x, x_n)$ and $d(x_n, y)$:

Rearrange to get $d(x, y) - d(x_n, y) \leq$

Pass 4 – Skeleton Proof **Homework 9, Problem 6 (continued)**

Finish.

Combine the two inequalities into an absolute value estimate:

Use squeeze theorem to conclude:

Use this page for the complete proof.

Official Solution (Professor)

Fix $y \in X$. By the triangle inequality,

$$\begin{aligned} d(x_n, y) \leq d(x_n, x) + d(x, y) \quad &\implies \quad d(x_n, y) - d(x, y) \leq d(x_n, x), \\ d(x, y) \leq d(x, x_n) + d(x_n, y) \quad &\implies \quad d(x, y) - d(x_n, y) \leq d(x_n, x). \end{aligned}$$

Together,

$$|d(x_n, y) - d(x, y)| \leq d(x_n, x).$$

Since $\{x_n\}_{n\geq 1}$ converges to x in (X, d), the right-hand side converges to 0. Therefore, by the squeeze theorem,

$$|d(x_n, y) - d(x, y)| \to 0,$$

and thus $d(x_n, y)$ converges to $d(x, y)$.

Reflection & Variation **Homework 9, Problem 6**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 10

Problem 1

Problem Statement **Homework 10, Problem 1**

Let (X, d) be a metric space and let $Y \subseteq X$. We call the set Y complete if the subspace Y with the induced metric is complete. Prove that if Y is complete, then Y is closed in X. (Hint: It suffices to show $Y \subseteq \overline{Y}$. Use sequences.)

Pass 1 – First Diagnosis **Homework 10, Problem 1**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card — Homework 10, Problem 1

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 10, Problem 1**

1. What does it mean for Y to be complete as a subspace of X?

---

2. What sequential characterization of closure should be used after fixing $x \in \overline{Y}$?

---

3. Why is a sequence converging in X automatically Cauchy in X?

---

4. Why is the same sequence Cauchy in the induced metric on Y?

---

5. Where is uniqueness of limits in a metric space used?

Pass 4 – Skeleton Proof **Homework 10, Problem 1**

Goal: prove Y is closed.
It suffices to show $\overline{Y} \subseteq Y$. Fix:

Use the sequential characterization of closure to get a sequence $\{y_n\} \subseteq Y$ such that:

Explain why $\{y_n\}$ is Cauchy in Y:

Use completeness of Y to get $y \in Y$ with:

Compare convergence in Y with convergence in X, then use uniqueness of limits:

Conclude $\overline{Y} \subseteq Y$, hence Y is closed:

Pass 5 – Full Proof Attempt	Homework 10, Problem 1

Use this page for the complete proof.

Student-Derived Solution (No official solution available)

We show that $\overline{Y} \subseteq Y$. Since always $Y \subseteq \overline{Y}$, this implies $\overline{Y} = Y$, and hence Y is closed in X.

Let $x \in \overline{Y}$. By the sequential characterization of closure, there exists a sequence $\{y_n\}_{n\geq 1}$ in Y such that

$$y_n \to x \qquad \text{in } (X, d).$$

Every convergent sequence in a metric space is Cauchy, so $\{y_n\}$ is Cauchy in (X, d). Since the induced metric on Y is the restriction of d, the same sequence is Cauchy in (Y, d_Y).

Because Y is complete, there is some $y \in Y$ such that

$$y_n \to y \qquad \text{in } (Y, d_Y).$$

Convergence in the subspace metric is the same distance condition as convergence in X, so $y_n \to y$ in (X, d). Thus the sequence $\{y_n\}$ converges to both x and y in X. By uniqueness of limits in metric spaces, $x = y$. Since $y \in Y$, we have $x \in Y$.

Therefore $\overline{Y} \subseteq Y$, so Y is closed in X.

Reflection & Variation **Homework 10, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 10, Problem 2**

Let (X, d) be a metric space. Prove that if $A, B \subseteq X$ are both compact then $A \cup B$ is compact.

Pass 1 – First Diagnosis **Homework 10, Problem 2**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 10, Problem 2

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 10, Problem 2**

1. What definition of compactness is most direct for a finite union?

---

2. Why does an open cover of $A \cup B$ automatically cover A and B separately?

---

3. What finite subcover do you get from compactness of A?

---

4. What finite subcover do you get from compactness of B?

---

5. Why is the union of the two finite index sets still finite and enough to cover $A \cup B$?

Pass 4 – Skeleton Proof **Homework 10, Problem 2**

Open-cover setup.
Let $\{U_i\}_{i\in I}$ be an arbitrary open cover of $A \cup B$:

Restrict the same cover to A, and use compactness of A to choose a finite $J_A \subseteq I$:

Restrict the same cover to B, and use compactness of B to choose a finite $J_B \subseteq I$:

Set $J =$

Check J is finite and $\{U_i\}_{i\in J}$ covers $A \cup B$:

Conclude compactness because the original open cover was arbitrary:

Use this page for the complete proof.

Student-Derived Solution (No official solution available)

Let $\{U_i\}_{i\in I}$ be an open cover of $A \cup B$. Thus each U_i is open in X, and

$$A \cup B \subseteq \bigcup_{i\in I} U_i.$$

Since $A \subseteq A \cup B$, the same family is an open cover of A. Because A is compact, there is a finite set $J_A \subseteq I$ such that

$$A \subseteq \bigcup_{i\in J_A} U_i.$$

Similarly, $\{U_i\}_{i\in I}$ is an open cover of B, so compactness of B gives a finite set $J_B \subseteq I$ such that

$$B \subseteq \bigcup_{i\in J_B} U_i.$$

Set $J = J_A \cup J_B$. Then J is finite, and

$$A \cup B \subseteq \left(\bigcup_{i\in J_A} U_i\right) \cup \left(\bigcup_{i\in J_B} U_i\right) = \bigcup_{i\in J} U_i.$$

Thus every open cover of $A \cup B$ has a finite subcover. Therefore $A \cup B$ is compact.

Reflection & Variation — Homework 10, Problem 2

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 10, Problem 3**

Let (X, d) be a metric space.

(a) Prove that if $A \subseteq B \subseteq X$, A is closed in X, and B is compact, then A is compact. (Hint: Given an open cover of A, we can add A^c to get an open cover of B.)

(b) Prove that if $C, D \subseteq X$ are both compact, then $C \cap D$ is compact. (Hint: Use (a).)

Pass 1 – First Diagnosis **Homework 10, Problem 3**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 10, Problem 3**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 10, Problem 3**

1. Why does adding A^c to an open cover of A help cover all of B?

---

2. Where is the assumption that A is closed used?

---

3. After compactness of B gives a finite subcover, why can A^c be discarded when covering A?

---

4. Which lecture result says compact subsets of a metric space are closed?

---

5. In part (b), which compact set plays the role of B in part (a)?

Pass 4 – Skeleton Proof **Homework 10, Problem 3**

Part (a): closed subset of a compact set.
Let $\{U_i\}_{i\in I}$ be an open cover of A. Since A is closed, A^c is:

Show $\{A^c\} \cup \{U_i\}_{i\in I}$ covers B:

Use compactness of B to extract a finite subcover:

Restrict back to A and remove the A^c part:

Part (b): compact intersection.
Use compactness $\Rightarrow$ closed to show $C \cap D$ is closed in X:

Apply part (a) with $A = C \cap D$ and $B =$

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Let $\{U_i\}_{i\in I}$ be an open cover of A. Since A is closed, A^c is open. Thus

$$\{A^c\} \cup \{U_i\}_{i\in I}$$

is an open cover of B, because

$$B \subseteq X = A^c \cup A \subseteq A^c \cup \bigcup_{i\in I} U_i.$$

Since B is compact, there is a finite subcover for B. We may write it as

$$\{A^c\} \cup \{U_i\}_{i\in J}$$

for some finite $J \subseteq I$. Then

$$B \subseteq A^c \cup \bigcup_{i\in J} U_i.$$

Since $A \subseteq B$ and $A \cap A^c = \varnothing$, it follows that

$$A \subseteq \bigcup_{i\in J} U_i.$$

Hence $\{U_i\}_{i\in J}$ is a finite subcover of A. Therefore A is compact.

(b) Since C and D are compact, they are closed by Lecture 27. Hence $C \cap D$ is closed. Also $C \cap D \subseteq C$, and C is compact. By part (a), $C \cap D$ is compact.

Reflection & Variation | **Homework 10, Problem 3**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 10, Problem 4**

Let (X, d) be a metric space and let $Y \subseteq X$. Prove that if Y is compact, then Y is complete.

Pass 1 – First Diagnosis **Homework 10, Problem 4**

Topic: ____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 10, Problem 4

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 10, Problem 4**

1. To prove completeness, what kind of sequence in Y should you start with?

2. Which compactness theorem gives a convergent subsequence in Y?

3. Why is a convergent subsequence not enough by itself to prove completeness?

4. How does the Cauchy property transfer convergence from a subsequence to the full sequence?

5. Why does the final limit lie in Y?

Pass 4 – Skeleton Proof **Homework 10, Problem 4**

Setup: prove every Cauchy sequence in Y converges in Y.
Let $\{y_n\}$ be a Cauchy sequence in Y:

Use compactness and Lecture 26 to obtain a subsequence $\{y_{k_n}\}$ and point $y \in Y$ such that:

Fix $r > 0$. From subsequence convergence choose N_1 such that:

From the Cauchy property choose N_2 such that:

Use triangle inequality with y_n, y_{k_n}, and y:

Conclude the whole sequence converges to $y \in Y$:

Use this page for the complete proof.

Official Solution (Professor)

Let $\{y_n\}_{n\geq 1}$ be a Cauchy sequence in Y. We show that it converges to some point of Y.
Since Y is compact, by Lecture 26 there exists a subsequence $\{y_{k_n}\}_{n\geq 1}$ that converges to some point $y \in Y$.
We prove the whole sequence converges to y. Fix $r > 0$. Since $y_{k_n} \to y$, there exists N_1 such that
$$d(y_{k_n}, y) < \frac{r}{2} \qquad \text{for all } n \geq N_1.$$
Since $\{y_n\}$ is Cauchy, there exists N_2 such that
$$d(y_n, y_m) < \frac{r}{2} \qquad \text{for all } n, m \geq N_2.$$
Set $N = \max\{N_1, N_2\}$. For $n \geq N$, we have $k_n \geq n \geq N$, and so
$$d(y_n, y) \leq d(y_n, y_{k_n}) + d(y_{k_n}, y) < \frac{r}{2} + \frac{r}{2} = r.$$
Since $r > 0$ was arbitrary, $y_n \to y$. Because $y \in Y$, every Cauchy sequence in Y converges in Y. Therefore Y is complete.

Reflection & Variation **Homework 10, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 11

Problem 1

Problem Statement **Homework 11, Problem 1**

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. Note that the set $\mathbb{Q}$ is not an interval, and so from lecture we know that it must be disconnected.

(a) Find two sets $A, B \subseteq \mathbb{R}$ that are nonempty and separated so that $\mathbb{Q} = A \cup B$, and prove your answer.

(b) Find a set $\varnothing \neq C \subsetneq \mathbb{Q}$ that is both open and closed in $\mathbb{Q}$, and prove your answer.

Pass 1 – First Diagnosis **Homework 11, Problem 1**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 11, Problem 1

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 11, Problem 1**

1. Why is cutting $\mathbb{Q}$ at an irrational number better than cutting it at a rational number?

---

2. What does it mean for two subsets of $\mathbb{R}$ to be separated?

---

3. What are the closures in $\mathbb{R}$ of $(-\infty, \sqrt{2}) \cap \mathbb{Q}$ and $(\sqrt{2}, \infty) \cap \mathbb{Q}$?

---

4. Why does $\sqrt{2} \notin \mathbb{Q}$ make the separation work?

---

5. How does the subspace characterization of open and closed sets in $\mathbb{Q}$ prove part (b)?

Pass 4 – Skeleton Proof **Homework 11, Problem 1**

Part (a): separated decomposition of $\mathbb{Q}$.
Choose an irrational cut point and define $A =$

Define $B =$

Check A and B are nonempty and $A \cup B = \mathbb{Q}$:

Compute $\overline{A}$ and $\overline{B}$ in $\mathbb{R}$:

Use those closures to prove $\overline{A} \cap B = \varnothing$ and $A \cap \overline{B} = \varnothing$:

Part (b): clopen subset of $\mathbb{Q}$.

Choose $C =$

Prove C is nonempty and proper:

Show C is open in $\mathbb{Q}$ and closed in $\mathbb{Q}$ using subspace results:

Use this page for part (a).

Use this page for part (b).

Student-Derived Solution (No official solution available)

(a) Define

$$A = (-\infty, \sqrt{2}) \cap \mathbb{Q}, \qquad B = (\sqrt{2}, \infty) \cap \mathbb{Q}.$$

Both sets are nonempty, for instance $1 \in A$ and $2 \in B$. Also $A \cup B = \mathbb{Q}$, since every rational number is either less than $\sqrt{2}$ or greater than $\sqrt{2}$, and $\sqrt{2} \notin \mathbb{Q}$. In $\mathbb{R}$,

$$\overline{A} = (-\infty, \sqrt{2}], \qquad \overline{B} = [\sqrt{2}, \infty).$$

Hence

$$\overline{A} \cap B = (-\infty, \sqrt{2}] \cap \Big((\sqrt{2}, \infty) \cap \mathbb{Q}\Big) = \varnothing,$$

and similarly

$$A \cap \overline{B} = \Big((-\infty, \sqrt{2}) \cap \mathbb{Q}\Big) \cap [\sqrt{2}, \infty) = \varnothing.$$

Therefore A and B are separated, nonempty, and satisfy $\mathbb{Q} = A \cup B$.

(b) Let

$$C = (-\infty, \sqrt{2}) \cap \mathbb{Q}.$$

Then $C \neq \varnothing$, since $1 \in C$, and $C \subsetneq \mathbb{Q}$, since $2 \in \mathbb{Q} \setminus C$.
The set $(-\infty, \sqrt{2})$ is open in $\mathbb{R}$, so $C = (-\infty, \sqrt{2}) \cap \mathbb{Q}$ is open in the subspace $\mathbb{Q}$. Also $(-\infty, \sqrt{2}]$ is closed in $\mathbb{R}$, and since $\sqrt{2} \notin \mathbb{Q}$,

$$(-\infty, \sqrt{2}] \cap \mathbb{Q} = (-\infty, \sqrt{2}) \cap \mathbb{Q} = C.$$

Thus C is closed in $\mathbb{Q}$. Therefore C is a nonempty proper subset of $\mathbb{Q}$ that is both open and closed in $\mathbb{Q}$.

Reflection & Variation **Homework 11, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 11, Problem 2**

Let (X, d) be a metric space.

(a) Prove that if $F \subseteq X$ is disconnected and closed in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.

(b) Prove that if $U \subseteq X$ is disconnected and open in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Pass 1 – First Diagnosis **Homework 11, Problem 2**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 11, Problem 2**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 11, Problem 2**

1. Which Lecture 29 characterization of disconnectedness gives a nontrivial clopen subset of F or U?

2. In part (a), why does closed in F plus F closed in X imply closed in X?

3. In part (b), why does open in U plus U open in X imply open in X?

4. Why are the two pieces nonempty?

5. How do closedness or openness of the pieces imply they are separated?

Pass 4 – Skeleton Proof **Homework 11, Problem 2**

Part (a): closed disconnected set.
Use disconnectedness of F to choose $A \neq \varnothing, F$ that is both open and closed in F. Set $B =$

Explain why A and B are nonempty and $F = A \cup B$:

Use F closed in X to prove A and B are closed in X:

Use closedness and disjointness to prove separatedness:

Part (b): open disconnected set.
Use disconnectedness of U to choose $A \neq \varnothing, U$ clopen in U, and set $B =$

Use U open in X to prove A and B are open in X:

Use complements of open sets to prove separatedness:

Use this page for part (a).

Pass 5 – Full Proof Attempt | Homework 11, Problem 2

Use this page for part (b).

Official Solution (Professor)

(a) Since F is disconnected, by Lecture 29 there exists a set $A \neq \varnothing, F$ that is both open and closed in F. Set $B = F \setminus A$.
The sets A and B are nonempty because $A \neq \varnothing$ and $A \neq F$. Since A is closed in F and F is closed in X, A is closed in X by Lecture 24. Also $B = F \setminus A$ is closed in F, because A is open in F; hence B is closed in X.
Finally, $A \cap B = \varnothing$. Since A and B are closed in X, $\overline{A} = A$ and $\overline{B} = B$. Thus

$$\overline{A} \cap B = \varnothing, \qquad A \cap \overline{B} = \varnothing.$$

Therefore A and B are nonempty, separated, closed in X, and $F = A \cup B$.

(b) Since U is disconnected, by Lecture 29 there exists a set $A \neq \varnothing, U$ that is both open and closed in U. Set $B = U \setminus A$.
The sets A and B are nonempty. Since A is open in U and U is open in X, A is open in X by Lecture 24. Since A is closed in U, $B = U \setminus A$ is open in U; because U is open in X, B is open in X.
Since $A \cap B = \varnothing$, we have $A \subseteq B^c$. Because B is open, B^c is closed, so $\overline{A} \subseteq B^c$, and hence $\overline{A} \cap B = \varnothing$. Similarly, $\overline{B} \cap A = \varnothing$. Therefore A and B are separated, open in X, nonempty, and $U = A \cup B$.

Reflection & Variation **Homework 11, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 11, Problem 3**

Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Pass 1 – First Diagnosis **Homework 11, Problem 3**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 11, Problem 3**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 11, Problem 3**

1. Why is contradiction the natural proof method here?

---

2. What does a separation of X give you?

---

3. Why can you choose one point a from one side and one point b from the other?

---

4. Which Lecture 29 result says a connected set inside a union of separated sets lies in one side?

---

5. Why does the connected set containing both a and b contradict the separation?

Pass 4 – Skeleton Proof **Homework 11, Problem 3**

Contradiction setup.
Assume X is disconnected. Then choose nonempty separated sets $A, B \subseteq X$ such that:

Choose $a \in A$ and $b \in B$. By hypothesis, choose connected $C \subseteq X$ such that:

Since $C \subseteq X = A \cup B$, apply the connected-set/separated-union result:

Use $a \in A \cap C$ and $b \in B \cap C$ to rule out both alternatives:

Conclude X is connected:

Pass 5 – Full Proof Attempt Homework 11, Problem 3

Use this page for the complete proof.

Official Solution (Professor)

Suppose toward a contradiction that X is disconnected. Then there exist sets $A, B \subseteq X$ such that

$$X = A \cup B, \qquad A \neq \varnothing, \qquad B \neq \varnothing,$$

and A and B are separated.

Choose $a \in A$ and $b \in B$. By the hypothesis, there exists a connected set $C \subseteq X$ such that $a, b \in C$. Since $C \subseteq X = A \cup B$, and A and B are separated, Lecture 29 implies that either $C \subseteq A$ or $C \subseteq B$.

Because $a \in A \cap C$, the only possible case is $C \subseteq A$. But then $B \cap C = \varnothing$, contradicting $b \in B \cap C$. Therefore X cannot be disconnected, so X is connected.

Reflection & Variation — Homework 11, Problem 3

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 11, Problem 4**

Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Pass 1 – First Diagnosis | Homework 11, Problem 4

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 11, Problem 4

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 11, Problem 4**

1. Why should the proof begin by assuming $\overline{A}$ is disconnected?

---

2. If $\overline{A} = B \cup C$ with B, C separated, what does connectedness of $A \subseteq B \cup C$ imply?

---

3. Why can one assume $A \subseteq B$ after relabeling?

---

4. How does $A \subseteq B$ imply $\overline{A} \subseteq \overline{B}$?

---

5. Why does separatedness then force the other side to be empty?

Pass 4 – Skeleton Proof **Homework 11, Problem 4**

Contradiction setup.
Assume $\overline{A}$ is disconnected, so $\overline{A} = B \cup C$ with B, C nonempty and separated:

Apply connectedness of A to the separated union $B \cup C$:

After relabeling, assume $A \subseteq B$. Then conclude $\overline{A} \subseteq$

Use separatedness $\overline{B} \cap C = \varnothing$ to show $\overline{A} \cap C = \varnothing$:

Explain why this contradicts $C \neq \varnothing$ and $C \subseteq \overline{A}$:

Use this page for the complete proof.

Official Solution (Professor)

Suppose toward a contradiction that $\overline{A}$ is disconnected. Then there exist sets $B, C \subseteq X$ such that

$$\overline{A} = B \cup C, \qquad B \neq \varnothing, \qquad C \neq \varnothing,$$

and B and C are separated.
Since A is connected and $A \subseteq \overline{A} = B \cup C$, Lecture 29 implies that either $A \subseteq B$ or $A \subseteq C$. After swapping B and C if necessary, assume $A \subseteq B$. Then

$$\overline{A} \subseteq \overline{B}.$$

Because B and C are separated, $\overline{B} \cap C = \varnothing$. Hence

$$\overline{A} \cap C = \varnothing.$$

But $C \subseteq \overline{A}$, since $\overline{A} = B \cup C$. Therefore $C = \varnothing$, a contradiction. Thus $\overline{A}$ is connected.

Reflection & Variation **Homework 11, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement **Homework 11, Problem 5**

Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \varnothing$, then $A \cup B$ is connected.

Pass 1 – First Diagnosis **Homework 11, Problem 5**

Topic: ______________________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 11, Problem 5

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 11, Problem 5**

1. Why does the assumption $A \cap B \neq \varnothing$ prevent A and B from lying in different sides of a separation?

---

2. If $A \cup B = C \cup D$ is a separation, what does connectedness of A imply?

---

3. What does connectedness of B imply?

---

4. How does a shared point force A and B into the same separated side?

---

5. Why does this make the other side empty?

Pass 4 – Skeleton Proof **Homework 11, Problem 5**

Contradiction setup.
Assume $A \cup B$ is disconnected. Choose nonempty separated $C, D \subseteq X$ such that:

Use connectedness of $A \subseteq C \cup D$ to conclude:

After relabeling, assume $A \subseteq C$. Use $A \cap B \neq \varnothing$ to show B has a point in C:

Use connectedness of B to conclude $B \subseteq C$:

Then $A \cup B \subseteq C$, contradicting:

Use this page for the complete proof.

Official Solution (Professor)

Suppose toward a contradiction that $A \cup B$ is disconnected. Then there exist sets $C, D \subseteq X$ such that

$$A \cup B = C \cup D, \qquad C, D \neq \varnothing,$$

and C and D are separated.

Since $A \subseteq C \cup D$ and A is connected, Lecture 29 implies that either $A \subseteq C$ or $A \subseteq D$. After swapping C and D, assume $A \subseteq C$.

Similarly, since $B \subseteq C \cup D$ and B is connected, either $B \subseteq C$ or $B \subseteq D$. But $A \subseteq C$ and $A \cap B \neq \varnothing$, so at least one point of B lies in C. Therefore B cannot be contained in D, and we must have $B \subseteq C$.

Thus $A \cup B \subseteq C$. Since $A \cup B = C \cup D$, this forces $D = \varnothing$, contradicting the separation. Therefore $A \cup B$ is connected.

Reflection & Variation **Homework 11, Problem 5**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement **Homework 11, Problem 6**

Let (X, d) be a metric space.

(a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.

(b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Pass 1 – First Diagnosis **Homework 11, Problem 6**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card **Homework 11, Problem 6**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 11, Problem 6**

1. What two closure-intersection conditions define separated sets?

---

2. In part (a), how do unions distribute over intersections with $\overline{A}$?

---

3. For part (b), why is contradiction a natural strategy?

---

4. If $F = A \cup B$ is a separation, how does connectedness of $F \cap G$ force $F \cap G$ into one side?

---

5. Why does proving B is separated from $A \cup G$ contradict connectedness of $F \cup G$?

Pass 4 – Skeleton Proof **Homework 11, Problem 6**

Part (a): separated from a union.
Write what it means that A is separated from B and from C:

Compute $\overline{A} \cap (B \cup C)$:

Use $\overline{B \cup C} = \overline{B} \cup \overline{C}$ to compute $A \cap \overline{B \cup C}$:

Conclude A and $B \cup C$ are separated:

Part (b): contradiction.
Assume F is disconnected and choose $F = A \cup B$ with A, B nonempty separated:

Use connectedness of $F \cap G$ to assume, after relabeling, $F \cap G \subseteq A$:

Show B and G are separated, using closedness of G:

Apply part (a) to show B and $A \cup G$ are separated, yielding a separation of $F \cup G$:

Use this page for part (a).

Use this page for part (b).

Official Solution (Professor)

(a) Since A and B are separated,

$$\overline{A} \cap B = \varnothing \qquad \text{and} \qquad A \cap \overline{B} = \varnothing.$$

Since A and C are separated,

$$\overline{A} \cap C = \varnothing \qquad \text{and} \qquad A \cap \overline{C} = \varnothing.$$

Therefore

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C) = \varnothing.$$

Also, by Lecture 23,

$$\overline{B \cup C} = \overline{B} \cup \overline{C}.$$

Hence

$$A \cap \overline{B \cup C} = (A \cap \overline{B}) \cup (A \cap \overline{C}) = \varnothing.$$

Thus A and $B \cup C$ are separated.

(b) Suppose toward a contradiction that F is disconnected. Then there exist $A, B \subseteq X$ such that

$$F = A \cup B, \qquad A, B \neq \varnothing,$$

and A and B are separated.
Since $F \cap G$ is connected and $F \cap G \subseteq A \cup B$, Lecture 29 implies that either $F \cap G \subseteq A$ or $F \cap G \subseteq B$. After swapping A and B if necessary, assume $F \cap G \subseteq A$. We claim that B and G are separated. First $B \cap G = \varnothing$, because any point in $B \cap G$ would lie in $F \cap G \subseteq A$, contradicting $A \cap B = \varnothing$. Since G is closed, $\overline{G} = G$, and this gives $B \cap \overline{G} = \varnothing$. Also, if $x \in G = \overline{G}$, then either $x \in G \setminus F$ or $x \in F \cap G \subseteq A$, so $x \notin B$. Hence $\overline{B} \cap G = \varnothing$. Thus B and G are separated.
Since A and B are separated, part (a) implies that B and $A \cup G$ are separated. Both are nonempty, and

$$F \cup G = (A \cup B) \cup G = (A \cup G) \cup B.$$

Therefore $F \cup G$ is disconnected, contradicting the hypothesis. Hence F is connected.

Reflection & Variation | **Homework 11, Problem 6**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Homework 12

Problem 1

Problem Statement **Homework 12, Problem 1**

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$, and $x_0 \in X$. Prove that f is continuous at x_0 if and only if for any open set $V \subseteq Y$ containing $f(x_0)$, there is an open set $U \subseteq X$ containing x_0 so that $x \in U$ implies $f(x) \in V$.

Pass 1 – First Diagnosis **Homework 12, Problem 1**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 12, Problem 1

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 12, Problem 1**

1. What is the ε-δ definition of continuity at x_0?

---

2. In the forward direction, how does openness of V produce a ball around $f(x_0)$?

---

3. Why is $U = B_\delta(x_0)$ the natural choice after applying continuity?

---

4. In the reverse direction, why should $V = B_\varepsilon(f(x_0))$ be used?

---

5. How does openness of U produce the δ needed for continuity?

Pass 4 – Skeleton Proof **Homework 12, Problem 1**

Forward direction.
Assume f is continuous at x_0. Fix open $V \subseteq Y$ with $f(x_0) \in V$. Choose $\varepsilon > 0$ so that:

Use continuity at x_0 to choose $\delta > 0$ so that:

Set $U =$

Check U is open, contains x_0, and $x \in U \Rightarrow f(x) \in V$:

Reverse direction.
Assume the open-neighborhood condition. Fix $\varepsilon > 0$ and set $V =$

Use the hypothesis to get open $U \ni x_0$ with $f(U) \subseteq V$:

Use openness of U to choose $\delta > 0$ with $B_\delta(x_0) \subseteq U$:

Conclude the ε-δ condition:

Pass 5 – Full Proof Attempt | Homework 12, Problem 1

Use this page for the forward direction.

Use this page for the reverse direction.

Student-Derived Solution (No official solution available yet)

$\Rightarrow$ Assume f is continuous at x_0. Let $V \subseteq Y$ be open with $f(x_0) \in V$. Since V is open, there is $\varepsilon > 0$ such that

$$B_\varepsilon(f(x_0)) \subseteq V.$$

By continuity at x_0, there is $\delta > 0$ such that

$$d_X(x, x_0) < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x_0)) < \varepsilon.$$

Let $U = B_\delta(x_0)$. Then U is open, $x_0 \in U$, and if $x \in U$, then $f(x) \in B_\varepsilon(f(x_0)) \subseteq V$.
$\Leftarrow$ Assume the open-set condition. Let $\varepsilon > 0$, and set $V = B_\varepsilon(f(x_0))$. Then V is open and contains $f(x_0)$, so there is an open set $U \subseteq X$ with $x_0 \in U$ and $f(U) \subseteq V$. Since U is open and $x_0 \in U$, there is $\delta > 0$ such that $B_\delta(x_0) \subseteq U$. Thus if $d_X(x, x_0) < \delta$, then $x \in U$, so $f(x) \in V$, meaning

$$d_Y(f(x), f(x_0)) < \varepsilon.$$

Therefore f is continuous at x_0.

Reflection & Variation **Homework 12, Problem 1**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 2

Problem Statement **Homework 12, Problem 2**

Let (X, d_X) and (Y, d_Y) be metric spaces and $f : X \to Y$ a function. Prove that f is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for any $A \subseteq X$.

Pass 1 – First Diagnosis **Homework 12, Problem 2**

Topic: ____________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 12, Problem 2

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 12, Problem 2**

1. What does $x \in \overline{A}$ mean sequentially in a metric space?

---

2. In the forward direction, how does continuity move a convergent sequence from X to Y?

---

3. Why does $f(a_n) \to f(x)$ prove $f(x) \in \overline{f(A)}$?

---

4. In the reverse direction, why choose $A = X \setminus f^{-1}(V)$ for an open $V \subseteq Y$?

---

5. Why does $x \notin \overline{A}$ produce an open neighborhood U of x with $f(U) \subseteq V$?

Pass 4 – Skeleton Proof **Homework 12, Problem 2**

Forward direction.
Assume f is continuous. Fix $A \subseteq X$ and $x \in \overline{A}$. Choose a sequence $a_n \in A$ such that:

Use continuity to show $f(a_n) \to$

Conclude $f(x) \in \overline{f(A)}$, hence $f(\overline{A}) \subseteq \overline{f(A)}$:

Reverse direction.
Assume $f(\overline{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$. Fix open $V \subseteq Y$ with $f(x) \in V$:

Define $A = X \setminus f^{-1}(V)$. Suppose for contradiction that $x \in \overline{A}$:

Use the closure hypothesis to force $f(x) \in \overline{f(A)}$:

Use $f(A) \subseteq Y \setminus V$ and $Y \setminus V$ closed to contradict $f(x) \in V$:

From $x \notin \overline{A}$, build an open $U \ni x$ with $U \subseteq f^{-1}(V)$, then apply Problem 1:

Use this page for the forward direction.

Use this page for the reverse direction.

Student-Derived Solution (No official solution available yet)

$\Rightarrow$ Assume f is continuous. Let $A \subseteq X$, and let $x \in \overline{A}$. By the sequential characterization of closure, there is a sequence $\{a_n\} \subseteq A$ such that $a_n \to x$. Since f is continuous,

$$f(a_n) \to f(x).$$

Each $f(a_n) \in f(A)$, so $f(x) \in \overline{f(A)}$. Therefore

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

$\Leftarrow$ Assume that for every $A \subseteq X$,

$$f(\overline{A}) \subseteq \overline{f(A)}.$$

We prove continuity using Problem 1. Let $V \subseteq Y$ be open and suppose $f(x) \in V$. Set

$$A = X \setminus f^{-1}(V).$$

We claim $x \notin \overline{A}$. If $x \in \overline{A}$, then by the hypothesis

$$f(x) \in f(\overline{A}) \subseteq \overline{f(A)}.$$

But if $a \in A$, then $f(a) \notin V$, so $f(A) \subseteq Y \setminus V$. Since $Y \setminus V$ is closed,

$$\overline{f(A)} \subseteq Y \setminus V.$$

This would imply $f(x) \notin V$, contradicting $f(x) \in V$. Thus $x \notin \overline{A}$.
Since $x \notin \overline{A}$, there exists an open set $U \subseteq X$ containing x such that $U \cap A = \varnothing$. Hence

$$U \subseteq X \setminus A = f^{-1}(V),$$

so $f(U) \subseteq V$. By Problem 1, f is continuous at x. Since $x \in X$ was arbitrary, f is continuous.

Reflection & Variation | **Homework 12, Problem 2**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 3

Problem Statement **Homework 12, Problem 3**

Let $f : \mathbb{R} \to \mathbb{R}$ be the function

$$f(x) = \frac{1}{1 + x^2}.$$

(a) Prove that f is continuous.

(b) Find an open set $U \subseteq \mathbb{R}$ so that $f(U)$ is not open.

(c) Find a closed set $F \subseteq \mathbb{R}$ so that $f(F)$ is not closed.

(d) Find a set $A \subseteq \mathbb{R}$ so that $f(\overline{A}) \neq \overline{f(A)}$.

Pass 1 – First Diagnosis **Homework 12, Problem 3**

Topic: ____

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 12, Problem 3

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 12, Problem 3**

1. Which standard limit laws prove continuity of $f(x) = 1/(1 + x^2)$?

---

2. What is the range of f on all of $\mathbb{R}$?

---

3. Why is $(0, 1]$ not open in $\mathbb{R}$?

---

4. Why is $(0, 1]$ not closed in $\mathbb{R}$?

---

5. For $A = (0, \infty)$, compare $f(\overline{A})$ with $\overline{f(A)}$. Where does the extra endpoint come from?

Pass 4 – Skeleton Proof | Homework 12, Problem 3

Part (a): continuity.
Use sequences or limit laws: if $x_n \to x$, then $x_n^2 \to$

Conclude $1 + x_n^2 \to 1 + x^2$, and since denominators stay nonzero:

Parts (b) and (c): image counterexamples.
For an open set, try $U = \mathbb{R}$. Compute $f(U) =$

Explain why that image is not open:

For a closed set, try $F = [0, \infty)$. Compute $f(F) =$

Explain why that image is not closed:

Part (d): closure/image mismatch.
Try $A = (0, \infty)$**. Then** $\overline{A} =$

Compute $f(\overline{A})$**,** $f(A)$**, and** $\overline{f(A)}$**:**

Use this page for parts (a) and (b).

Use this page for parts (c) and (d).

Student-Derived Solution (No official solution available yet)

(a) Let $x_n \to x$ in $\mathbb{R}$. Then by the standard limit laws,

$$x_n^2 \to x^2, \qquad 1 + x_n^2 \to 1 + x^2.$$

Since $1 + x^2 \neq 0$, the reciprocal limit law gives

$$\frac{1}{1 + x_n^2} \to \frac{1}{1 + x^2}.$$

Hence $f(x_n) \to f(x)$, so f is continuous.

(b) Let $U = \mathbb{R}$, which is open. Then

$$f(U) = (0, 1].$$

This is not open in $\mathbb{R}$, since $1 \in (0, 1]$ but no open interval around 1 is contained in $(0, 1]$.

(c) Let $F = [0, \infty)$, which is closed in $\mathbb{R}$. Then

$$f(F) = (0, 1].$$

This set is not closed in $\mathbb{R}$, since 0 is a limit point of $(0, 1]$ but $0 \notin (0, 1]$.

(d) Let $A = (0, \infty)$. Then $\overline{A} = [0, \infty)$, so

$$f(\overline{A}) = f([0, \infty)) = (0, 1].$$

On the other hand,

$$f(A) = f((0, \infty)) = (0, 1), \qquad \overline{f(A)} = [0, 1].$$

Therefore

$$f(\overline{A}) = (0, 1] \neq [0, 1] = \overline{f(A)}.$$

Reflection & Variation **Homework 12, Problem 3**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 4

Problem Statement **Homework 12, Problem 4**

Consider the metric spaces $(\mathbb{R}, |\cdot|)$ and $(\mathbb{R}^2, d_2)$.

(a) Prove that the functions $f, g : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x, y) = x$ and $g(x, y) = y$ are continuous. (Hint: Use sequences.)

(b) Prove that if $h : [0, 1] \to \mathbb{R}^2$ is a continuous function such that $h(0) = (x_0, y_0)$, $h(1) = (x_1, y_1)$, and $x_0 < 0 < x_1$, then there exists a point $t \in (0, 1)$ so that $h(t)$ lies on the y-axis.

Pass 1 – First Diagnosis **Homework 12, Problem 4**

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 12, Problem 4

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 12, Problem 4**

1. How does convergence in $(\mathbb{R}^2, d_2)$ imply coordinatewise convergence?

---

2. Why does $|x_n - x| \leq d_2((x_n, y_n), (x, y))$ prove continuity of the first projection?

---

3. What scalar function should be formed from the path h in part (b)?

---

4. Why is $\phi(t) = f(h(t))$ continuous?

---

5. How does the Intermediate Value Theorem produce a point on the y-axis?

Pass 4 – Skeleton Proof **Homework 12, Problem 4**

Part (a): projections are continuous.
Let $(x_n, y_n) \to (x, y)$ in $(\mathbb{R}^2, d_2)$. Write the coordinate inequality for x_n:

Use squeeze to prove $x_n \to x$, hence $f(x_n, y_n) \to f(x, y)$:

Repeat the coordinate inequality for y_n:

Part (b): path crosses the y-axis.
Define $\phi : [0, 1] \to \mathbb{R}$ by $\phi(t) =$

Explain why ϕ is continuous:

Compute $\phi(0)$ and $\phi(1)$, and record their signs:

Apply the Intermediate Value Theorem to get $t \in (0, 1)$ with $\phi(t) = 0$:

Translate $\phi(t) = 0$ into "$h(t)$ lies on the y-axis":

Use this page for part (a).

Use this page for part (b).

Student-Derived Solution (No official solution available yet)

(a) We prove continuity of $f(x, y) = x$. Let $(x_n, y_n) \to (x, y)$ in $(\mathbb{R}^2, d_2)$. Then

$$d_2((x_n, y_n), (x, y)) = \sqrt{(x_n - x)^2 + (y_n - y)^2} \to 0.$$

Since

$$|x_n - x| \leq d_2((x_n, y_n), (x, y)),$$

the squeeze theorem gives $x_n \to x$. Thus

$$f(x_n, y_n) = x_n \to x = f(x, y),$$

so f is continuous. The proof for $g(x, y) = y$ is the same: use

$$|y_n - y| \leq d_2((x_n, y_n), (x, y))$$

to get $y_n \to y$, hence $g(x_n, y_n) \to g(x, y)$.

(b) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the first-coordinate projection from part (a), and define

$$\phi : [0, 1] \to \mathbb{R}, \qquad \phi(t) = f(h(t)).$$

Since h is continuous and f is continuous, $\phi = f \circ h$ is continuous. Also

$$\phi(0) = f(h(0)) = f(x_0, y_0) = x_0 < 0,$$

and

$$\phi(1) = f(h(1)) = f(x_1, y_1) = x_1 > 0.$$

By the Intermediate Value Theorem, there exists $t \in (0, 1)$ such that $\phi(t) = 0$. This means the first coordinate of $h(t)$ is 0, so $h(t) \in \{(0, y) : y \in \mathbb{R}\}$. Therefore $h(t)$ lies on the y-axis.

Reflection & Variation **Homework 12, Problem 4**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 5

Problem Statement **Homework 12, Problem 5**

Going forward, you may assume (no proof necessary) any fact about $\sin x$ and $\cos x$ from calculus, such as $\sin, \cos : \mathbb{R} \to [-1, 1]$ are continuous functions.
Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = x^2 + y^2 + \sin(x - y)$$

has a global minimum. (Hint: Notice that we cannot directly apply the extreme value theorem here. However, we can apply it to any compact subset of $\mathbb{R}^2$.)

Pass 1 – First Diagnosis **Homework 12, Problem 5**

Topic: ______________________

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card Homework 12, Problem 5

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 12, Problem 5**

1. Why can the extreme value theorem not be applied directly on all of $\mathbb{R}^2$?

---

2. Why is f continuous?

---

3. What lower bound follows from $\sin(x - y) \geq -1$?

---

4. Why does the closed unit disk capture a global minimizer?

---

5. Why is it useful to find one point inside the disk where $f < 0$?

Pass 4 – Skeleton Proof **Homework 12, Problem 5**

Continuity and coercive estimate.
Show f is continuous as a sum/composition of continuous functions:

Use $\sin(x-y) \geq -1$ **to prove** $f(x,y) \geq$

Conclude that if $x^2+y^2 \geq 1$**, then** $f(x,y) \geq$

Find a compact region where the minimum occurs.
Choose $p = (-1/2, 1/2)$**. Compute** $f(p) =$

Explain why $f(p) < 0$**:**

Let $K = \{(x,y) : x^2+y^2 \leq 1\}$**. Explain why** K **is compact:**

Apply the extreme value theorem to $f|_K$:

Compare points inside K and outside K to prove the minimizer on K is global:

Pass 5 – Full Proof Attempt Homework 12, Problem 5

Use this page for the setup and compact restriction.

Pass 5 – Full Proof Attempt Homework 12, Problem 5

Use this page to prove the minimum is global.

Student-Derived Solution (No official solution available yet)

First, f is continuous on $\mathbb{R}^2$. The map $(x, y) \mapsto x^2 + y^2$ is continuous, the map $(x, y) \mapsto x - y$ is continuous, and sin is continuous by the allowed calculus facts. Hence $(x, y) \mapsto \sin(x - y)$ is continuous, and so f is continuous.
Since $\sin(x - y) \geq -1$, we have

$$f(x, y) = x^2 + y^2 + \sin(x - y) \geq x^2 + y^2 - 1.$$

In particular, if $x^2 + y^2 \geq 1$, then $f(x, y) \geq 0$.
Now consider

$$p = \left(-\frac{1}{2}, \frac{1}{2}\right).$$

Then

$$f(p) = \frac{1}{4} + \frac{1}{4} + \sin(-1) = \frac{1}{2} - \sin(1) < 0.$$

Also p lies in the closed unit disk

$$K = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

The set K is closed and bounded in $\mathbb{R}^2$, hence compact. By the extreme value theorem, since f is continuous, there exists $q \in K$ such that

$$f(q) \leq f(z) \qquad \text{for all } z \in K.$$

We claim q is a global minimizer on $\mathbb{R}^2$. If $z \in K$, this is true by the choice of q. If $z \notin K$, then $z = (x, y)$ satisfies $x^2 + y^2 > 1$, so $f(z) \geq 0$. But

$$f(q) \leq f(p) < 0 \leq f(z).$$

Thus $f(q) \leq f(z)$ for every $z \in \mathbb{R}^2$. Therefore f has a global minimum.

Reflection & Variation **Homework 12, Problem 5**

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:

Problem 6

Problem Statement **Homework 12, Problem 6**

Let X and Y be metric spaces and $f : X \to Y$ a function. Prove that if X is compact and f is continuous and bijective, then the inverse function $f^{-1} : Y \to X$ is also continuous. (Hint: It suffices to show that the preimage of any closed subset of X under f^{-1} is closed in Y.)

Pass 1 – First Diagnosis Homework 12, Problem 6

Topic: ______

What is being asked?:

Definitions likely relevant:

Theorem/proof method likely relevant:

What should the first line of the proof be?:

Pass 2 – Proof Recipe Card | **Homework 12, Problem 6**

Hypotheses:

Conclusion:

Definitions to unpack:

Useful theorem/lemma:

Key inequality/identity:

One-sentence proof strategy:

Pass 3 – Self-Explanation **Homework 12, Problem 6**

1. Which closed-set criterion for continuity is suggested by the hint?

---

2. If $C \subseteq X$ is closed, why is C compact?

---

3. Why is $f(C)$ compact in Y?

---

4. Why is $f(C)$ closed in Y?

---

5. Why is $(f^{-1})^{-1}(C) = f(C)$ when f is bijective?

Pass 4 – Skeleton Proof **Homework 12, Problem 6**

Use the closed-set criterion for f^{-1}.
Let $C \subseteq X$ be closed. Since X is compact, conclude C is:

Since f is continuous, conclude $f(C)$ is:

Since Y is a metric space, compact subsets of Y are:

Use bijectivity to compute $(f^{-1})^{-1}(C) =$

Conclude the preimage under f^{-1} of every closed set $C \subseteq X$ is closed in Y:

Apply the closed-set criterion to finish:

Pass 5 – Full Proof Attempt Homework 12, Problem 6

Use this page for the complete proof.

Student-Derived Solution (No official solution available yet)

We prove that $f^{-1} : Y \to X$ is continuous using the closed-set criterion.
Let $C \subseteq X$ be closed. Since X is compact and C is closed in X, C is compact. Since $f : X \to Y$ is continuous, the image of a compact set under f is compact. Therefore $f(C)$ is compact in Y.
Because Y is a metric space, compact subsets of Y are closed. Hence $f(C)$ is closed in Y.
Since f is bijective,

$$(f^{-1})^{-1}(C) = f(C).$$

Thus the preimage under f^{-1} of every closed subset $C \subseteq X$ is closed in Y. By the closed-set criterion for continuity, f^{-1} is continuous.

Reflection & Variation	Homework 12, Problem 6

What was the key definition?:

What was the key proof move?:

What step would I forget under time pressure?:

What nearby exam variation could appear?:
